CISCaRL

Detailed Model Creation Report

31 steps, with the exact output of every step

Support-constrained interpretable subgroup discovery for survival outcomes

Dataset: ACTG175, arms 0 and 1, n = 1,054

Final model: C200-I · horizon: 730 days

Prepared: November 2026

Every result exhibit is the output of the executed notebook, copied verbatim. The figures are the saved paper figures generated from the result files.

Project at a glance

Dataset	ACTG175, arms 0 and 1, n = 1,054; 12 baseline covariates
Estimand	τ(730|x) = S1(730|x) − S0(730|x), composite progression-free endpoint
Partitions	527 discovery / 263 validation / 264 assessment (disjoint)
Estimator	Per-arm Cox + conditional censoring Cox, 5-fold cross-fit, AIPW
Learner	Depth-3 CART, minimum leaf 60, squared error
Selection	Highest discovery mean with support ≥ 200; fallback flagged
Decision	Treat only if the validation one-sided 95% lower bound is above zero
Final ACTG175 result	Rule support 61; validation LCB −0.250; decision INSUFFICIENT

How to read this report

The report follows the 31 steps of the executed notebook. Each step begins with the code that was used, then explains what was done and why it matters. The result paragraph gives the numbers in plain language. The exhibit that follows is the exact output of that step, copied verbatim from the notebook; where a figure exists, it is shown as well. The final paragraph says what the result means and what decision it forced.

What is shown for every step

- Code used — the script or notebook cell that produced the result.
- What we did — the action in plain language.
- Why this step matters — the risk it controls.
- Result obtained — the key numbers in one short paragraph.
- Exact result — a picture of the console output from the executed notebook, plus any saved figure.
- What this means — the inference, including what is not supported.
- Decision at this step — the choice that was frozen or rejected.

The complete document is long by design: it is meant to be read beside the notebook or used as an audit trail. Every number can be found in results/ and every exhibit can be rebuilt by running the notebook or run_all.py.

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Step 1 — Environment and reproducibility

Code used: ciscarl2/audit.py, requirements.txt

What we did – Before touching the data, the project fixes the Python version and every package, loads the data path, and runs an audit. The audit checks small, hand-verifiable things: a rule's conditions must match the patients it selects, tree leaves must not overlap, the influence-function contrast must equal the direct formula, and the purity, capture and Jaccard definitions must match hand counts. It also simulates a tiny dataset with a known answer to check the AIPW estimator.

Why this step matters – Every later number depends on the estimator and the rule code being correct. The audit makes that assumption testable instead of assumed, and it is cheap to run.

Result obtained: All checks pass. AIPW bias is +0.0031 with a standard deviation of 0.0130, and the two arm means are within 0.002 of the truth.

What this means: The building blocks are correct before any result is quoted. If this audit failed, the pipeline would stop here; because it passes, the environment and seeds can be frozen for the rest of the study.

Decision at this step: Freeze the environment and seeds, and proceed only because the audit passes.

Exact result

```
python : 3.11.0
numpy  : 2.4.6
pandas : 2.2.3
root   : C:\Users\roshn\OneDrive\Desktop\Amrita\SEM 5\ML\Project
data   : C:\Users\roshn\OneDrive\Desktop\Amrita\SEM 5\ML\Project\data\ACTG175.csv

=== CISCaRL implementation audit ===

[PASS] extract_leaf_rules: condition == mask  max diff 0
[PASS] extract_leaf_rules: leaves partition the data

[PASS] assign: most-specific rule wins, default elsewhere
[PASS] paired IF contrast D and SE  D=-0.026925 vs -0.026925
[PASS] purity/capture/Jaccard definitions

      AIPW bias +0.0031 (sd 0.0130), IF-CI coverage 100%
      arm means: phi1 - S1 +0.0015, phi0 - S0 -0.0016
[PASS] AIPW bias < 0.01
[PASS] arm means within 0.01

all core checks passed

audit failures: []
```

Exhibit 1. Exact console output of Step 1, copied verbatim from the executed notebook.

Step 2 — Study the dataset

Code used: notebook Step 2 (pandas profiling)

What we did – The raw file has 27 columns. Each column gets a role: identifier, outcome, treatment, post-baseline, design, or baseline covariate. The project then checks missingness, variance, redundancy, sparsity and implausible values, so the covariate set is chosen from evidence rather than from habit.

Why this step matters – A subgroup model fails quietly when an input is constant, duplicated, measured after treatment, or missing for most patients. This step decides the 12 inputs and records why each other column was excluded.

Result obtained: Fifteen of the 27 columns are not baseline covariates. Of the 16 baseline candidates, four are removed: zprior is constant in arms 0/1, z30 agrees with str2 in 94.8% of patients, oprior is present in only 2.4%, and preanti is a redundant prior-treatment duration. cd496 is missing for 797 of 2,139 patients and is post-baseline. Three patients have cd40 = 0, 199 fall outside the trial's 200–500 range, and 13 are minors.

What this means: This step produces the 12 baseline covariates used everywhere later. Each exclusion has a reason, so the model is not silently dropping difficult patients or keeping a duplicate variable.

Decision at this step: Keep the 12 baseline covariates; exclude the other 15 columns with recorded reasons.

Exact result

```
raw shape: (2139, 27)
 pidnum  age   wtkg  hemo  homo  drugs  karnof  oprior  z30  zprior  ...  offtrt  cd40  cd420  cd496  r  cd80  cd820  cens  days  arms
  10056   48 89.813     0     0      0     100       0    0       1  ...       0   422    477  660.0  1   566    324     0   948     2
  10059   61 49.442     0     0      0      90       0    1       1  ...       0   162    218    NaN  0   392    564     1  1002     3
  10089   45 88.452     0     1      1      90       0    1       1  ...       1   326    274  122.0  1  2063   1893     0   961     3
  10093   47 85.277     0     1      0     100       0    1       1  ...       0   287    394    NaN  0  1590    966     0  1166     3
  10124   43 66.679     0     1      0     100       0    1       1  ...       0   504    353  660.0  1   870    782     0  1090     0

27-variable role and screening audit
variable                      meaning              role                                screening decision  missing  n_unique
  pidnum           patient identifier        identifier             not a patient characteristic  exclude        0      2139
     age                 age (years)          baseline                                 suitable   retain        0        59
    wtkg                 weight (kg)          baseline                                 suitable   retain        0       667
    hemo          haemophilia status          baseline                                 suitable   retain        0         2
    homo         homosexual activity          baseline                                 suitable   retain        0         2
   drugs        intravenous drug use          baseline                                 suitable   retain        0         2
  karnof  Karnofsky performance score          baseline                                 suitable   retain        0         4
  oprior     prior antiretroviral use baseline candidate                      2.4% prevalence  exclude        0         2
     z30          prior AZT > 30 days baseline candidate                94.8% agreement with str2  exclude        0         2
  zprior         prior antiretroviral baseline candidate                zero variance in arms 0/1  exclude        0         1
 preanti prior antiretroviral duration baseline candidate                  redundant with str2/z30  exclude        0       813
```

Exhibit 2. Exact console output of Step 2 (part 1 of 3), copied verbatim from the executed notebook.

```
    race                        race          baseline                                  suitable   retain        0         2
  gender                      gender          baseline                                  suitable   retain        0         2
    str2      antiretroviral history          baseline                                  suitable   retain        0         2
   strat       randomisation stratum            design                derived design variable  exclude        0         3
 symptom          symptomatic status          baseline                                  suitable   retain        0         2
   treat         treatment indicator         treatment                       duplicates arms  exclude        0         2
  offtrt   off treatment during study      post-baseline                 post-baseline event  exclude        0         2
    cd40          baseline CD4 count          baseline                                  suitable   retain        0       484
   cd420              CD4 at week 20      post-baseline          measured after randomisation  exclude        0       570
   cd496              CD4 at week 96      post-baseline measured after randomisation; 68% missing  exclude      797       534
       r       progression indicator            outcome                         outcome-related  exclude        0         2
    cd80          baseline CD8 count          baseline                                  suitable   retain        0      1090
   cd820              CD8 at week 20      post-baseline          measured after randomisation  exclude        0      1050
    cens  event / censoring indicator            outcome                   target variable (E)  exclude        0         2
    days              follow-up time            outcome                   target variable (T)  exclude        0       713
    arms               treatment arm         treatment                 exposure variable (A)  exclude        0         4

decision counts
decision
exclude    15
retain     12
```

Exhibit 3. Exact console output of Step 2 (part 2 of 3), copied verbatim from the executed notebook.

```
retained features (12):
age, wtkg, hemo, homo, drugs, karnof, race, gender, str2, symptom, cd40, cd80

model-ready matrix: 1054 rows x 12 features

treatment arms: {0: 532, 1: 522, 2: 524, 3: 561}

arms 0/1: (1054, 27)
zero-variance columns: ['zprior']
  z30      agreement with str2: 0.948 | prevalence 0.549
  zprior   agreement with str2: 0.586 | prevalence 1.000
  oprior   agreement with str2: 0.437 | prevalence 0.024

implausible values (arms 0/1):
  cd40 == 0      : 3
  cd40 outside 200-500: 199
  age < 18       : 13
  karnof < 70    : 0
```

Exhibit 4. Exact console output of Step 2 (part 3 of 3), copied verbatim from the executed notebook.

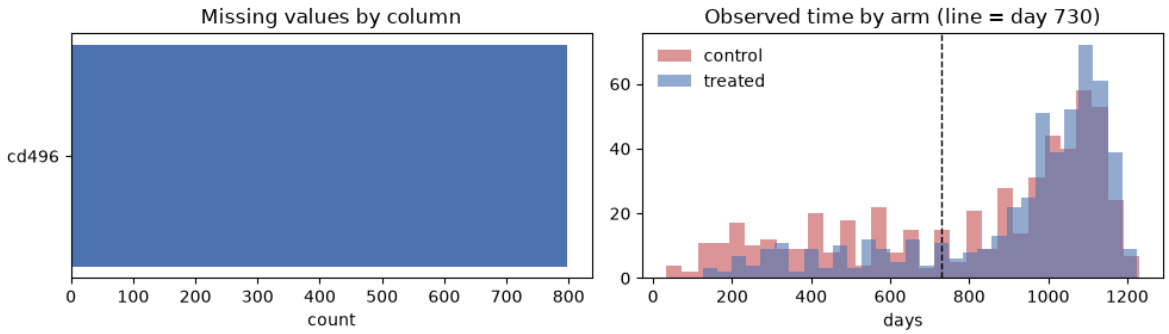


Exhibit 5. Missing values by column and observed follow-up time by arm. Only cd496 has missing values; the dashed line marks 730 days.

Step 3 — Cohort and outcome definition

Code used: notebook Step 3 (cohort and endpoint construction)

What we did – The analysis sample, time origin, composite event and horizon are fixed. The composite event is a CD4 decline of at least 50, AIDS progression, or death. The project then checks that the horizon is inside the observed follow-up and that enough patients are still being followed at that point.

Why this step matters – The endpoint and time origin define what 'benefit' means. A different endpoint or a different start time would answer a different clinical question.

Result obtained: 1,054 of 2,139 patients remain (522 treated and 532 control). There are 284 composite events (26.9%), 103 patients are censored before day 730 (9.8%), and 750 (71.2%) are still at risk at the horizon. Median observed time is 989 days, maximum 1,231.

What this means: The horizon is observable rather than extrapolated. Censoring before 730 days is modest but real, so the estimator must adjust for it. The 284 events are the binding limit on subgroup precision, which is why accuracy is measured in simulation.

Decision at this step: Use the composite progression-free event and the 730-day horizon, with censoring adjustment.

Exact result

```
analysis cohort: 1054 | treated 522 | control 532
events: 284 (26.9%)
censored before 730: 103 (9.8%)
at risk at 730: 750 (71.2%)
median observed time: 989.0
```

Exhibit 6. Exact console output of Step 3, copied verbatim from the executed notebook.

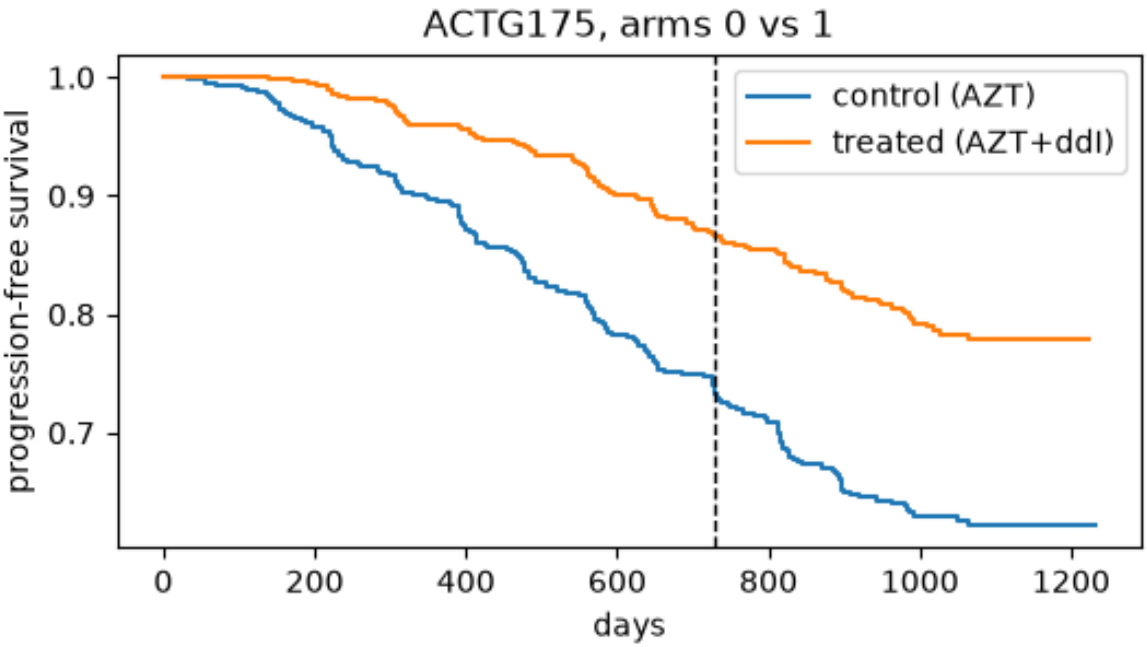


Exhibit 7. Kaplan–Meier curves for the two treatment arms on ACTG175, with 730 days marked.

Step 4 — Split the data

Code used: notebook Step 4 (train_test_split; seeds 2024/2025)

What we did – The cohort is divided into discovery, validation and assessment, stratified by treatment and event together. The row indices are frozen, so the same patients stay in the same role for the whole project.

Why this step matters – If the same patient appears in discovery and validation, every later comparison is optimistic. The split is the structural defence against that.

Result obtained: The split is 527 / 263 / 264, with event rates 27.1% / 26.6% / 26.9% — a spread of 0.5 points, compared with 2.8 points when stratifying on treatment alone. A separate ratio study found 50/25/25 gave the highest capture (0.479) at purity 0.941 with 80% commitment. Every ratio produced 20 distinct rules from 20 splits.

What this means: Validation and assessment are small, so an uneven event rate would add noise to every later comparison; joint stratification removes that noise. The 50/25/25 ratio is kept because it balances discovery size and validation precision. The rule instability is recorded now, not hidden.

Decision at this step: Fix the 527/263/264 split, stratified by treatment and event together.

Exact result

```
 partition   n  treated  events  event_rate
 discovery 527      261     143       0.271
validation 263      130      70       0.266
assessment 264      131      71       0.269
disjoint: True

REAL DATA - ACTG175, 20 splits per ratio
   split  discovery  validation  median_support  median_val_SE  median_LCB  commit_pct  distinct_rules
40/30/30        421         316            67.5          0.147      -0.081        30.0              20
50/25/25        527         263            77.0          0.157      -0.087        25.0              20
60/20/20        632         211           207.5          0.126      -0.110         5.0              20
70/15/15        737         158           231.0          0.142      -0.090        30.0              20

SIMULATION - single region, n = 8000, 20 replications per ratio
   split  purity  capture    mae  commit_pct  coverage  median_support
40/30/30  0.8677   0.3777 0.0611        80.0    0.1055           319.0
50/25/25  0.9408   0.4790 0.0590        80.0    0.1297           653.5
60/20/20  0.9404   0.4322 0.0589        65.0    0.1123           666.0
70/15/15  0.9650   0.4633 0.0585        60.0    0.1169           853.0
```

Exhibit 8. Exact console output of Step 4, copied verbatim from the executed notebook.

Step 5 — Choose the horizon

Code used: ciscarl2/estimators.py, notebook Step 5

What we did – Several candidate horizons are evaluated under pre-declared constraints: at least 150 events, at least 60% of patients still at risk, and at most 15% censoring. The horizon is not chosen by picking the largest estimated effect.

Why this step matters – A horizon chosen because the estimated effect is largest is selected on the outcome. The horizon must be fixed by follow-up constraints before the effect is read.

Result obtained: The minimum standard error is at 365 days, the maximum signal-to-noise ratio is around 900 days, and 730 days sits between them with 71% of the cohort at risk and under 10% censored. The constraints admit roughly 550–900 days, and 730 is fixed inside that range as the two-year window. A sensitivity analysis covers 365 and 1,095 days.

What this means: Choosing the horizon where the effect looks largest would be selection on the outcome. 730 days is a defensible compromise inside a range that satisfies precision constraints, and the sensitivity analysis shows how much the answer moves at the edges.

Decision at this step: Fix 730 days inside the feasible range; report 365 and 1,095 days as sensitivity analyses.

Exact result

```
   t   tau    se  tau_over_se  events  at_risk  censored
 365 0.062 0.029        2.109      46    0.875     0.038
 456 0.095 0.032        2.943      55    0.844     0.051
 548 0.121 0.036        3.399      69    0.806     0.063
 640 0.155 0.040        3.892      84    0.755     0.085
 730 0.180 0.043        4.135      99    0.708     0.104
 821 0.200 0.046        4.335     115    0.662     0.120
 912 0.215 0.053        4.076     135    0.579     0.165
1003 0.172 0.069        2.480     141    0.454     0.279
1095 0.261 0.146        1.782     143    0.218     0.510
min SE at t = 365
max tau/SE at t = 821
```

Exhibit 9. Exact console output of Step 5, copied verbatim from the executed notebook.

Step 6 — Preprocessing fitted on discovery

Code used: ciscarl2/estimators.py, notebook Step 6

What we did – The scaler is fitted on the discovery partition only. The step then checks whether standardisation is actually required by the Cox prediction formula, and tests common transforms such as logarithms and winsorising.

Why this step matters – Scaling on all patients leaks information, and the Cox prediction formula assumes centred covariates. Fitting the scaler on discovery only is both a leakage control and a correctness requirement.

Result obtained: Discovery means differ from full-cohort means (cd80 1,019.5 vs 995.7; drugs 0.112 vs 0.129), which proves the scaler never saw validation or assessment. With raw covariates the predicted control survival at 730 days is 0.909; with standardised covariates it is 0.732, a maximum per-patient difference of 0.43. Log transformation and winsorising change nothing once the model is fitted.

What this means: Standardisation is a correctness requirement, not a convenience. The prediction formula assumes centred covariates, and the nuisance model is weakly penalised, so scale affects the fit. Fitting the scaler on discovery only keeps the validation and assessment partitions untouched.

Decision at this step: Fit the scaler on discovery only and standardise every covariate.

Exact result

```
covariate  discovery_mean  discovery_sd  full_mean  full_sd
     cd40         349.313       121.957    350.986  122.245
     cd80        1019.539       505.811    995.702  481.195
      age          35.528         8.952     35.228    8.769
     wtkg          75.299        13.790     75.472   13.423
   karnof          95.503         5.883     95.484    5.869
     hemo           0.078         0.268      0.081    0.272
     homo           0.649         0.477      0.652    0.476
     race           0.269         0.444      0.279    0.448
   gender           0.824         0.381      0.822    0.383
    drugs           0.112         0.315      0.129    0.335
     str2           0.598         0.490      0.586    0.492
  symptom           0.178         0.383      0.176    0.380
discovery means == full means: False

mean S(730) on control arm:
  raw covariates      : 0.909
  standardised        : 0.737
  max |raw - centred| : 0.424
```

Exhibit 10. Exact console output of Step 6, copied verbatim from the executed notebook.

Step 7 — Leakage checklist

Code used: notebook Step 7 (mechanical leakage checks)

What we did – This step proves mechanically that no validation or assessment information entered any fitted quantity. It checks the covariate list, the scaler means and the partition membership.

Why this step matters – Leakage is the most common reason a model looks good in development and fails in use. A mechanical checklist is stronger than an intention to be careful.

Result obtained: No post-baseline or outcome column is in X. The scaler means equal the discovery means and differ from the full-cohort means. The three partitions share no patient and have the intended sizes.

What this means: The mechanical leakage channels are closed. The remaining risk is analyst-side: looking at assessment before the analysis is frozen. That is controlled by the order of the notebook, not by code, which is why the notebook separates discovery, validation and assessment so strictly.

Decision at this step: Pass the leakage checklist and keep validation and assessment untouched.

Exact result

```
                         check          result
     post-baseline columns in X              []
           outcome columns in X              []
scaler means == discovery means            True
     scaler means == full means           False
            partitions disjoint            True
                partition sizes 527 / 263 / 264
```

Exhibit 11. Exact console output of Step 7, copied verbatim from the executed notebook.

Step 8 — Nuisance estimation

Code used: ciscarl2/estimators.py (crossfit_nuisances)

What we did – Two nuisance models are fitted on discovery: a survival model for each treatment arm and a conditional censoring model. Both use five-fold cross-fitting, so a patient's own outcome never helps predict that patient. Censoring weights are then capped to stop a few extreme values from dominating.

Why this step matters – The effect estimate needs a good outcome model and honest censoring weights. Cross-fitting stops a patient from helping predict their own outcome.

Result obtained: Fold sizes are 106/106/105/105/105. Weight caps range from 1.23 to 1.35. Of 527 patients, 373 carry a positive weight; the maximum weight is 1.335 and the 99th percentile is 1.31. Effective sample size is 372 (71% of discovery), and 10.4% of discovery patients are censored before day 730.

What this means: The weights barely bind, so the estimate rests on observed follow-up rather than on extrapolation. Cross-fitting is still required because the nuisances are fitted on the same patients they later score. An effective sample size of 372 is the honest denominator for precision claims.

Decision at this step: Use per-arm Cox plus a conditional censoring Cox, five folds, and cap weights at min(Q0.99, 20).

Exact result

quantity	value
fold sizes	[106, 106, 105, 105, 105]
cap min	1.225
cap max	1.348
weights > 0	373
max weight	1.335
p99 weight	1.308
effective sample size	372.2
censored before 730	0.104

Exhibit 12. Exact console output of Step 8, copied verbatim from the executed notebook.

Step 9 — Target construction (AIPW)

Code used: ciscarl2/estimators.py, ciscarl2/gate2.py (pseudo_outcome)

What we did – The two nuisance models are combined into one pseudo-outcome per patient. This number is the AIPW estimate of that patient's treatment effect: it uses the outcome model but also corrects it with the observed outcome, so it stays consistent if one of the models is wrong.

Why this step matters – The tree needs one number per patient. The choice of target determines what the tree can find, and a noisy target makes discovery fail even when a real subgroup exists.

Result obtained: The pseudo-outcome is centred at +0.180 with a standard deviation of 0.997. The audit showed AIPW bias +0.0031 (SD 0.0130) and arm means within 0.002. The global proportional-hazards test does not reject (p = 0.235 and 0.368), but cd40 violates in both arms (p = 0.021, 0.023). Efron and Breslow tie handling differ by at most 0.0014. Calibration at 730 days has a weighted mean absolute gap of 0.032.

What this means: The pseudo-outcome is the target the tree needs: one number per patient whose average is the treatment effect, already adjusted for censoring. Its spread near 1 is the heterogeneity signal; if it were tightly concentrated, no rule could beat the constant. The cd40 violation is a limitation, not a reason to abandon the model, because the global test holds.

Decision at this step: Use the AIPW pseudo-outcome as the discovery target; record the cd40 proportional-hazards violation.

Exact result

```
psi mean : 0.1797  (AIPW marginal estimate)
psi sd   : 0.9967
AIPW ATE : 0.1797

Proportional-hazards test (rank transform), global Fisher p per arm:
 arm  statistic  p_value
   0     28.610    0.235
   1     25.709    0.368

Covariates with p < 0.05:
 arm test  p_value
   0 cd40    0.021
   0 str2    0.022
   1 cd40    0.023
   1 wtkg    0.048

Cox ties: Efron vs Breslow coefficients
 arm  max_abs_coef_diff  mean_abs_coef_diff  corr
   0             0.0005              0.0001   1.0
   1             0.0014              0.0004   1.0
```

Exhibit 13. Exact console output of Step 9 (part 1 of 2), copied verbatim from the executed notebook.

```
Survival calibration at 730 days (bin 0 = whole arm)
 arm  bin   n  predicted  observed_km    gap
   0    0 133      0.738        0.732 -0.006
   0    1  45      0.552        0.572  0.020
   0    2  44      0.775        0.695 -0.080
   0    3  44      0.891        0.929  0.038
   1    0 131      0.872        0.859 -0.013
   1    1  44      0.773        0.689 -0.084
   1    2  43      0.893        0.974  0.081
   1    3  44      0.951        0.923 -0.028

Weight-cap sensitivity
 cap  pct_capped  selected_leaf  support  mean_psi  fallback  max_weight_raw     ess
 1.2       9.867              6       71     0.485      True           1.402 526.338
 1.5       0.000              2       61     0.556      True           1.402 525.975
 2.0       0.000              2       61     0.556      True           1.402 525.975
 5.0       0.000              2       61     0.556      True           1.402 525.975
20.0       0.000              2       61     0.556      True           1.402 525.975
```

Exhibit 14. Exact console output of Step 9 (part 2 of 2), copied verbatim from the executed notebook.

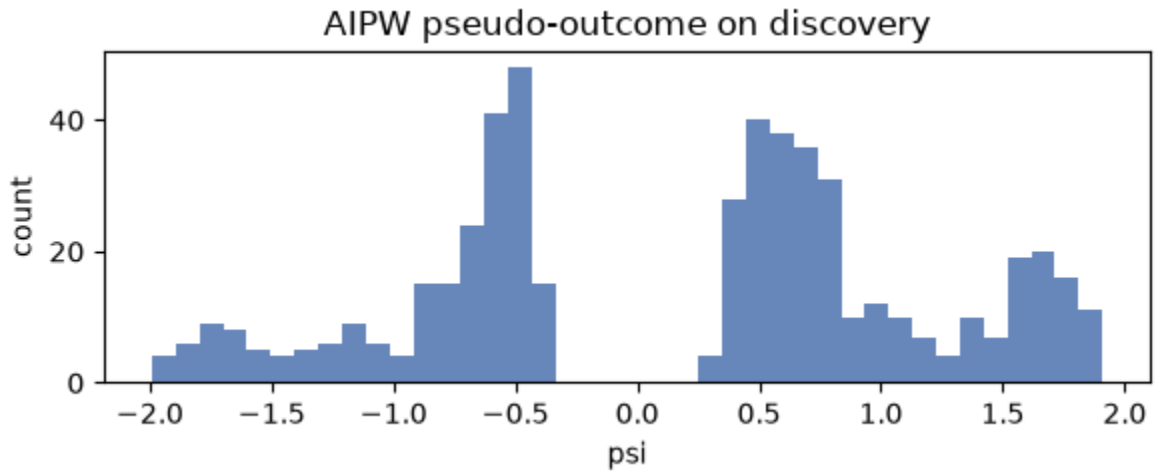


Exhibit 15. Distribution of the AIPW pseudo-outcome on the discovery partition. The spread is the heterogeneity signal the tree uses.

Step 10 — Rule discovery (decision tree)

Code used: ciscarl2/rules.py, ciscarl2/gate11.py (_cart)

What we did – A small regression tree is fitted to the pseudo-outcome. Each root-to-leaf path becomes a candidate rule, such as 'cd40 ≤ 407.5 AND wtkg > 71.0'. Depth is capped at three so a rule has at most three conditions and can be read aloud.

Why this step matters – The rule must be simple enough to read and specific enough to matter. Depth is an interpretability constraint, not a tuning knob.

Result obtained: The tree produces six leaves with supports from 60 to 173 patients and mean pseudo-outcomes from −0.184 to +0.556. Two leaves are negative and one is strongly positive.

What this means: The tree separates a high-benefit subgroup from a negative one — exactly the heterogeneity the method exists to find. Two of six leaves being negative also shows that 'treat the best leaf' is not the same as 'treat everyone above average'. Depth 3 is fixed for interpretability; leaf size is tuned later.

Decision at this step: Use depth 3 and treat every root-to-leaf path as a candidate rule.

Exact result

leaf	rule	support	mean_psi
1	cd40 <= 407.5 & wtkg <= 71.0 & homo <= 0.5	71	0.282
2	cd40 <= 407.5 & wtkg > 71.0 & homo <= 0.5	61	0.556
3	cd40 <= 238.5 & homo > 0.5	67	-0.094
4	cd40 <= 407.5 & cd40 > 238.5 & homo > 0.5	173	0.228
5	cd40 > 407.5 & cd80 <= 977.0	60	-0.184
6	cd40 > 407.5 & cd80 > 977.0	95	0.196

Exhibit 16. Exact console output of Step 10, copied verbatim from the executed notebook.

Step 11 — Selection and commitment

Code used: ciscarl2/estimators.py, notebook Step 11

What we did – The support floor is applied: a leaf must contain at least 200 discovery patients to be selected. The chosen rule is then re-estimated on validation, and treatment is recommended only if the one-sided 95% lower confidence bound is above zero.

Why this step matters – The best leaf in discovery is often a small lucky leaf. The support floor and the validation bound decide whether that leaf should be trusted.

Result obtained: No leaf reaches 200 patients (the largest has 173), so the fallback is used and leaf 2 is selected: cd40 ≤ 407.5 & wtkg > 71.0 & homo ≤ 0.5, with 61 discovery patients. On validation (n = 31) the effect is +0.116 (SE 0.222), giving a lower bound of −0.250 and the decision INSUFFICIENT. On assessment the rule shows +0.401 [+0.033, +0.770] on 38 patients, but that is descriptive only.

What this means: The support floor cannot bind at 527 discovery patients, and the validation partition is too small to certify a 61-patient leaf. The model therefore refuses to recommend treatment. That is the commitment layer doing its job, not a bug.

Decision at this step: Apply support 200 and the validation lower-bound rule; abstain when the bound is below zero.

Exact result

```
         quantity                                        value
             leaf                                            2
             rule cd40 <= 407.5 & wtkg > 71.0 & homo <= 0.5
discovery support                                           61
   support >= 200                                        False
         fallback                                         True
     validation n                                           31
   validation tau                                        0.116
              SE                                         0.222
             LCB                                         -0.25
          commit                                        False

validation: n=31 | tau=+0.116 | SE=0.222 | LCB=-0.250
decision: INSUFFICIENT

Assessment is not read here. It is computed once in Step 23, after the
rule and the decision procedure are frozen.
```

Exhibit 17. Exact console output of Step 11, copied verbatim from the executed notebook.

Step 12 — Simulation harness

Code used: ciscarl2/dgp.py

What we did – Real data cannot tell us whether a subgroup is correct. The project therefore builds semi-synthetic trials from the real ACTG175 covariate matrix, generates survival and

censoring times, and adds a known treatment effect in one or two regions. Because the effect is generated, every patient's true benefit is known.

Why this step matters – Real data have no individual treatment-effect truth, so accuracy cannot be measured without simulation. The simulator supplies the missing answer.

Result obtained: The harness gives 25.5% prevalence, a 20.0% event rate, 10.1% censoring, a true average effect of +0.050, and a true effect of +0.195 inside the region and 0 outside.

What this means: The truth is analytic, so accuracy, purity and capture can be computed honestly. The marginal effect is only +0.05 because most patients do not benefit; that dilution is why a model predicting the average everywhere cannot win, and why capture and purity are the informative metrics.

Decision at this step: Use the semi-synthetic ACTG175-based simulator for every accuracy claim.

Exact result

```
prevalence of the true region: 0.255
event rate by 730           : 0.2
censored before 730         : 0.101
true ATE                    : 0.05
true tau inside region      : 0.195
true tau outside region     : 0.0
```

Exhibit 18. Exact console output of Step 12, copied verbatim from the executed notebook.

Step 13 — First result: the constant baseline

Code used: ciscarl2/gate17.py, results/gate17_summary.csv

What we did – The rule model is compared with the simplest possible policy: give everyone the same marginal effect. The comparison is repeated at 2,000, 4,000 and 6,000 patients, with 40 seeds per cell. An oracle model that knows the true effect is included as an upper bound.

Why this step matters – A subgroup model is only useful if it beats the simple policy of treating everyone at the average effect. Without this baseline, a gain can be imaginary.

Result obtained: At n = 2,000 the rule model is worse than the constant (−26.4%, only 17.5% of replications better). At 4,000 it is marginal (+4.5%, 55% better). At 6,000 it is clearly better (+21.7%, 85% better). The oracle sits 40–60 points higher throughout.

What this means: The method has a measured operating range, not a universal advantage. Below roughly 5,000 patients it should not be used. The gap to the oracle is a discovery gap, not an information gap: the signal exists, but a depth-3 tree cannot recover it at small sample sizes.

Decision at this step: Record the operating range; do not present the model as universally better than the constant.

Exact result

n	target	mae	const_mae	rel_gain	frac_better
2000	oracle	0.04	0.07	40.12	85.0
2000	psi	0.09	0.07	-26.44	17.5
4000	oracle	0.03	0.07	59.50	100.0
4000	psi	0.07	0.07	4.53	55.0
6000	oracle	0.03	0.07	61.34	100.0
6000	psi	0.06	0.07	21.74	85.0

Exhibit 19. Exact console output of Step 13, copied verbatim from the executed notebook.

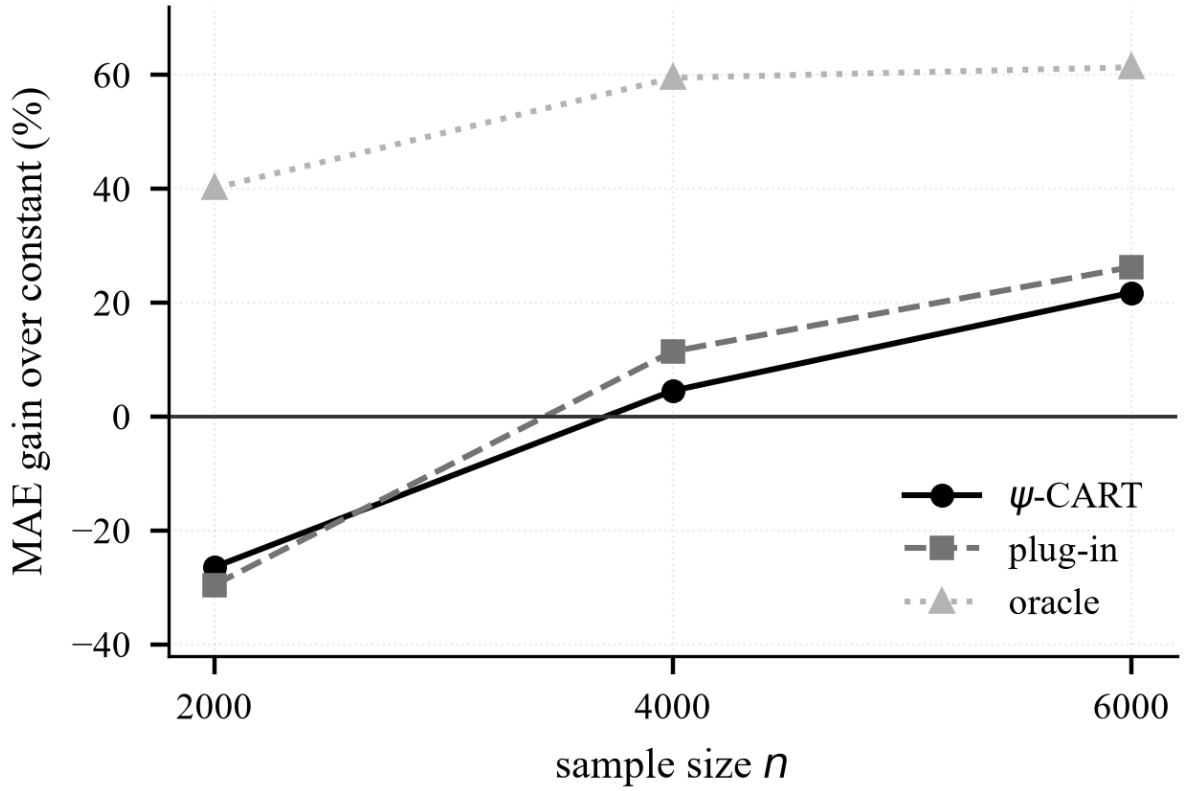


Exhibit 20. Operating range: rule model versus the constant baseline, with the oracle bound.

Step 14 — Support floor

Code used: ciscarl2/gate21.py, gate23.py, gate30.py

What we did – The minimum subgroup size is swept to measure the trade-off between purity (how many treated patients truly benefit) and capture (how much of the total benefit the rule finds). The sweep includes no floor, 100, 200 and 400 patients, and a proportional floor at small sample sizes.

Why this step matters – Small leaves win by luck. The support floor is the main defence against the winner's curse, and its cost in purity must be measured.

Result obtained: In the two-region design, no floor gives capture 0.149 and purity 0.774; a floor of 100 gives 0.246/0.847; 200 gives 0.364/0.820; 400 gives 0.441/0.778. In the single-region design, no floor gives 0.228/0.626 and a floor of 200 gives 0.514/0.821. At n = 2,000 the

absolute floor gives purity 0.47 and commitment 17.5%, while a 5% proportional floor gives 0.73 and 30%.

What this means: The floor roughly triples capture at a purity cost of about 0.02 in the two-region design, and improves purity in the single-region design. A floor of 400 buys more capture but gives purity back, so 200 is the better operating point. At small n the absolute floor is disproportionate, which is a documented limitation.

Decision at this step: Adopt the 200-patient floor; keep 400 as a sensitivity.

Exact result

```
two-region design, floor sweep (40 seeds):
config  support  capture  purity  j_true  commit  coverage  per_treated
     A  247.975    0.148   0.774   0.127    45.0     0.040        0.173
     B  310.900    0.194   0.826   0.166    55.0     0.057        0.175
  C100  433.025    0.246   0.847   0.201    62.5     0.084        0.170
  C200  677.450    0.364   0.820   0.295    70.0     0.129        0.163
  C400  916.175    0.441   0.778   0.345    75.0     0.162        0.158
     D  565.525    0.210   0.668   0.179    52.5     0.084        0.131

single-region, max-mean vs support >= 200:
arm  support  capture  purity  jaccard  commit_rate  coverage
  A  284.600    0.228   0.626    0.216         42.5     0.051
  B  418.725    0.351   0.781    0.328         60.0     0.085
  C  649.700    0.514   0.821    0.462         80.0     0.142
  D  577.050    0.229   0.414    0.203         35.0     0.049

small samples: absolute 200 vs proportional 5%
   n variant  purity  capture  commit  fallback
1000  abs200   0.614    0.105    27.5      80.0
1000   prop5   0.624    0.102    27.5       0.0
2000  abs200   0.468    0.096    17.5       0.0
2000   prop5   0.728    0.127    30.0       0.0
4000  abs200   0.710    0.311    55.0       0.0
4000   prop5   0.804    0.281    55.0       0.0
```

Exhibit 21. Exact console output of Step 14, copied verbatim from the executed notebook.

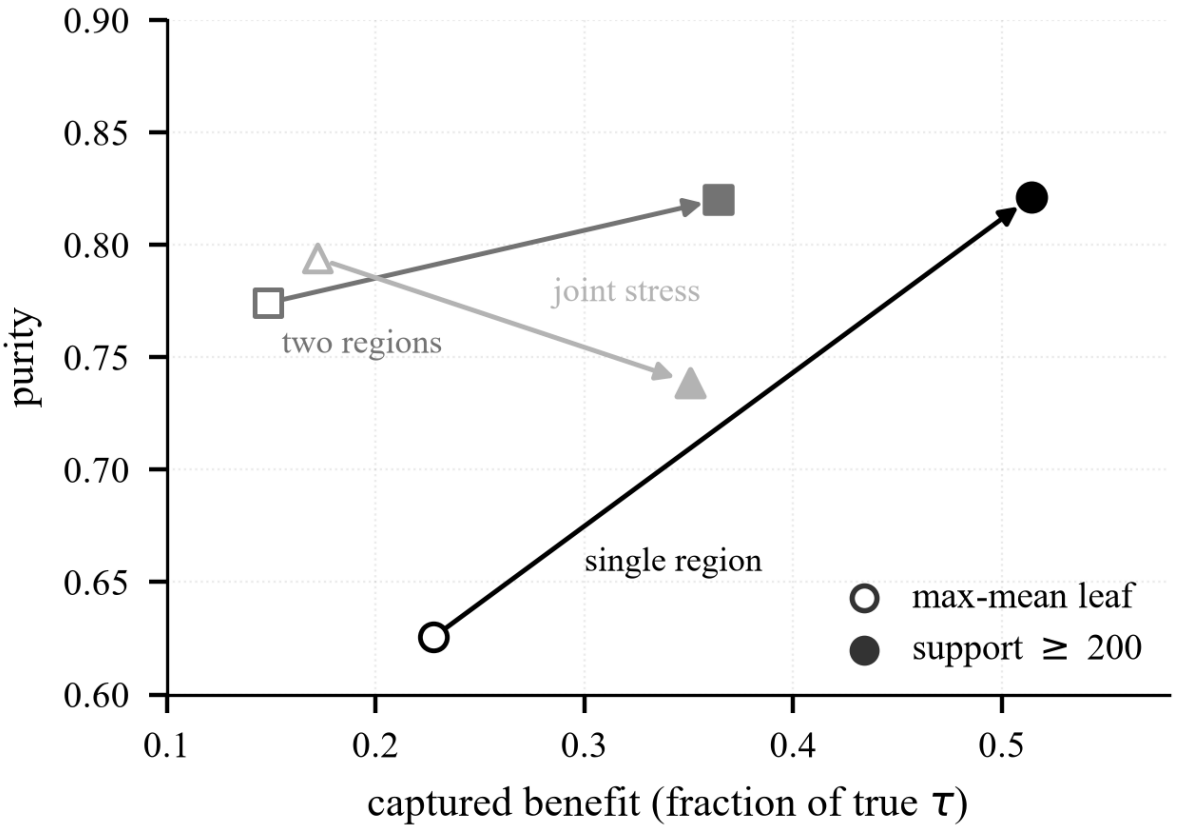


Exhibit 22. Purity–capture frontier for the support floor.

Step 15 — Calibration

Code used: ciscarl2/gate30.py (shrinkage variants)

What we did – Leaf effects are noisy because each leaf contains only a few dozen patients. Calibration pulls those noisy effects toward the overall average. The project compares no calibration with empirical-Bayes shrinkage, using 80 paired replications across two designs.

Why this step matters – Leaf effects are noisy because each leaf contains only a few dozen patients. Calibration changes reported effects without changing who is treated.

Result obtained: Shrinkage lowers MAE by 0.0051 [−0.0077, −0.0025] in the single-region design and 0.0026 [−0.0048, −0.0005] in the two-region design. Purity, capture, commitment and rank correlation are unchanged. The head-to-head with SCRE moves from a significant deficit to a tie.

What this means: The pattern is pure variance reduction: the leaf means are noisy, and pulling them toward the average improves prediction without changing who is treated. Because this was chosen after the SCRE comparison, it is reported as a post-hoc calibration (C200-S), confirmed on fresh seeds, not merged into the frozen configuration.

Decision at this step: Adopt empirical-Bayes shrinkage as a post-hoc prediction calibration (C200-S) and confirm it on fresh seeds.

Exact result

```
variant sweep, pooled over 80 paired replications (negative = better):
  kind variant   mae  mae_base  d_mae     lo     hi     p
single      V2 0.052     0.058 -0.005 -0.008 -0.002 0.000
single     V3a 0.058     0.058 -0.000 -0.000  0.000 0.159
single     V3b 0.058     0.058  0.000  0.000  0.001 0.016
single     V4a 0.058     0.058 -0.000 -0.001  0.001 0.762
single     V4b 0.058     0.058  0.000 -0.001  0.002 0.788
single      V5 0.068     0.058  0.010  0.008  0.012 0.000
   two      V2 0.068     0.071 -0.003 -0.005 -0.000 0.014
   two     V3a 0.070     0.071 -0.000 -0.000  0.000 0.131
   two     V3b 0.071     0.071  0.000  0.000  0.001 0.019
   two     V4a 0.071     0.071  0.000 -0.001  0.001 0.675
   two     V4b 0.069     0.071 -0.002 -0.003 -0.001 0.000
   two      V5 0.079     0.071  0.008  0.007  0.010 0.000

policy quantities are unchanged by shrinkage (confirmation block):
  kind method   mae  scre_mae  purity  capture  coverage
single   C200 0.058     0.051   0.890    0.465     0.133
single C200-S 0.053     0.051   0.890    0.465     0.133
   two   C200 0.071     0.068   0.858    0.277     0.117
   two C200-S 0.068     0.068   0.858    0.277     0.117
```

Exhibit 23. Exact console output of Step 15, copied verbatim from the executed notebook.

Step 16 — Nuisance and estimator comparison

Code used: ciscarl2/gate31.py, gate36.py, gate39.py

What we did – Three alternatives are tested against the frozen choices: flexible outcome models (random survival forest, gradient boosting, equal-weight stacking), a fixed-horizon TMLE estimator, and learned stacking of Cox, RSF and GBM. Everything else is held fixed.

Why this step matters – A flexible outcome model may fit survival better but produce worse treatment effects. The comparison must be on the effect, not on the survival fit alone.

Result obtained: Cox has the best Brier score (0.0028 single, 0.0033 two). Stacking lowers patient-level MAE (0.0475/0.0504 vs 0.0517/0.0579) but loses 0.030 purity in the two-region design and costs 136–260 s per replication vs 0.002 s. GBM loses 0.036 purity. TMLE is purer (+0.064/+0.068) but its MAE is 0.089/0.080 worse and Brier is 0.015/0.014 worse. The learned stack puts about 0.77 weight on Cox and is indistinguishable from Cox (+0.0007/+0.0001 MAE).

What this means: Cox and AIPW are retained. A flexible model buys a small MAE gain and pays with purity, calibration and a very large cost. TMLE is not a patient-level improvement. The result is a tested negative result for complexity, not an unexamined preference for the incumbent.

Decision at this step: Keep Cox and AIPW; reject RSF, GBM, equal-weight stacking, learned stacking and fixed-horizon TMLE.

Exact result

```
  kind model  reps   bias  coverage  mae_nuis  brier  purity  capture  commit  seconds
single   cox    20 -0.004      90.0     0.052  0.003   0.794    0.623    75.0    0.002
single   rsf    20 -0.004      85.0     0.054  0.007   0.802    0.615    70.0  119.816
single   gbm    20 -0.004      85.0     0.053  0.007   0.795    0.640    70.0  147.467
single stack    20 -0.004      90.0     0.047  0.004   0.793    0.623    75.0  260.428
   two   cox    20 -0.004      85.0     0.058  0.003   0.833    0.424    80.0    0.001
   two   rsf    20 -0.004      85.0     0.056  0.007   0.820    0.445    80.0   77.212
   two   gbm    20 -0.004      85.0     0.055  0.007   0.797    0.400    75.0   60.962
   two stack    20 -0.004      85.0     0.050  0.005   0.803    0.433    80.0  136.355

Paired vs Cox (Holm on MAE):
  kind model  d_mae_nuis  lo_mae_nuis  hi_mae_nuis  p_holm  d_brier  d_purity  d_capture
single   rsf       0.002        0.001        0.004   0.001    0.005     0.008     -0.008
single   gbm       0.001       -0.000        0.003   0.189    0.004     0.000      0.017
single stack      -0.004       -0.006       -0.003   0.000    0.002    -0.001     -0.000
   two   rsf      -0.002       -0.004       -0.000   0.012    0.004    -0.013      0.021
   two   gbm      -0.003       -0.005       -0.002   0.001    0.004    -0.036     -0.023
   two stack      -0.008       -0.009       -0.006   0.000    0.001    -0.030      0.009

  kind estimator   bias  coverage  mae_nuis  brier  purity  capture  commit  mae_assessment  seconds
single      aipw -0.001      95.0     0.052  0.003   0.869    0.582    70.0           0.062    0.000
single      tmle -0.005     100.0     0.141  0.018   0.932    0.542    85.0           0.052    4.549
```

Exhibit 24. Exact console output of Step 16 (part 1 of 2), copied verbatim from the executed notebook.

```
   two      aipw -0.002      95.0     0.058  0.003   0.836    0.390    80.0           0.070    0.000
   two      tmle -0.011     100.0     0.137  0.017   0.904    0.398    90.0           0.066    4.672

Paired TMLE vs AIPW:
  kind  d_mae_nuis  lo_mae_nuis  hi_mae_nuis  p_mae_nuis  d_brier  d_purity  d_capture
single       0.089        0.088        0.091         0.0    0.015     0.064     -0.039
   two       0.080        0.078        0.081         0.0    0.014     0.068      0.008

  kind estimator  mae_nuis  purity  capture  commit  mae_assessment  w_cox  w_rsf  w_gbm
single       cox     0.051   0.844    0.560    80.0           0.052  0.771  0.135  0.094
single     equal     0.047   0.885    0.524    80.0           0.056  0.771  0.135  0.094
single     learn     0.051   0.869    0.551    80.0           0.054  0.771  0.135  0.094
   two       cox     0.057   0.817    0.413    80.0           0.069  0.756  0.228  0.016
   two     equal     0.051   0.817    0.413    90.0           0.069  0.756  0.228  0.016
   two     learn     0.057   0.817    0.413    90.0           0.069  0.756  0.228  0.016

Paired vs Cox:
  single equal dMAE=-0.0038 [-0.0048,-0.0027] dPurity=+0.041 dCapture=-0.036
  single learn dMAE=+0.0007 [-0.0008,+0.0028] dPurity=+0.024 dCapture=-0.009
     two equal dMAE=-0.0064 [-0.0074,-0.0054] dPurity=+0.000 dCapture=+0.000
     two learn dMAE=+0.0001 [-0.0016,+0.0022] dPurity=+0.000 dCapture=+0.000
```

Exhibit 25. Exact console output of Step 16 (part 2 of 2), copied verbatim from the executed notebook.

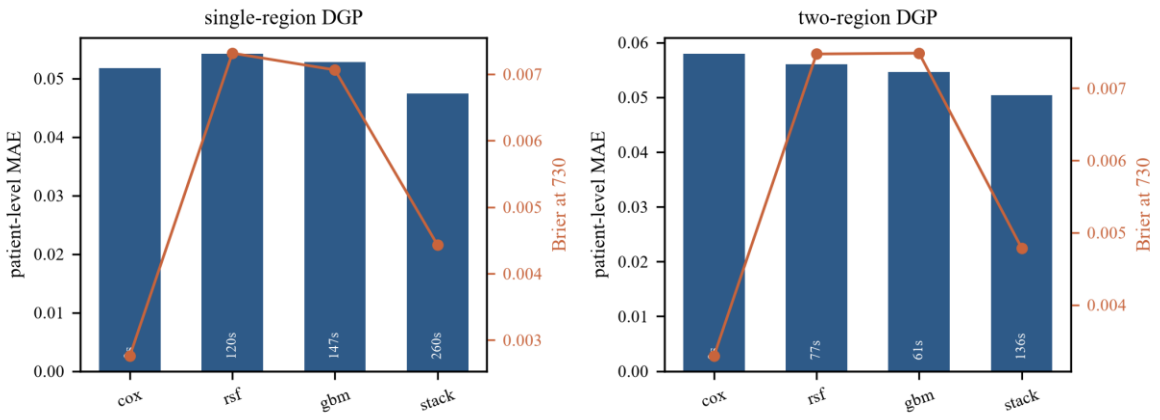


Exhibit 26. Gate 31: Cox versus random survival forest, gradient boosting and equal-weight stacking.

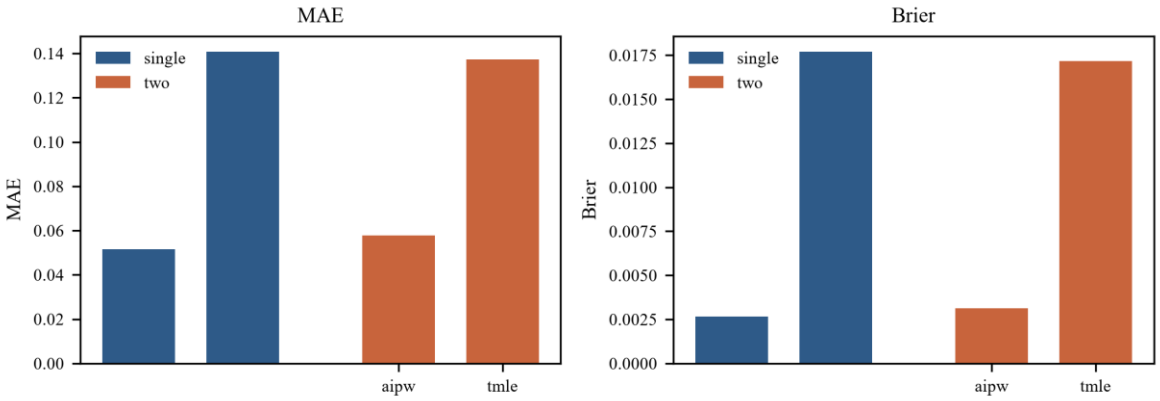


Exhibit 27. Gate 36: AIPW versus fixed-horizon IPCW-TMLE.

Step 17 — Rule learners

Code used: ciscarl2/gate32.py

What we did – Five learners are compared on the same pseudo-outcome and the same selection protocol: regression CART, cost-complexity-pruned CART, PRIM peeling boxes, a classification tree on the sign of the effect, and boosting distillation.

Why this step matters – Different learners produce different rules. The simplest learner has to be tested against the alternatives rather than assumed to be best.

Result obtained: Pruned CART gains capture in the two-region design (+0.261) but loses 0.173 purity, nearly nine times the allowed loss of 0.02. The classification tree loses 0.219/0.110 purity. PRIM loses both capture and purity and needs 4.6–4.8 conditions. Boosting is the only learner with a better two-region MAE (−0.0029) but loses capture (−0.092) and purity (−0.050) and costs 25–105 s per replication.

What this means: No alternative meets the pre-registered guardrails. Regression CART is kept. Higher capture is not worth a rule that treats many people who do not benefit.

Decision at this step: Keep regression CART; reject pruned CART, PRIM, the classification tree and boosting distillation.

Exact result

```
  kind learner  purity  capture  jaccard  commit  mae_assessment  support  conditions  seconds
single    cart   0.889    0.531    0.487    80.0           0.056   305.05        2.80    0.062
single  pruned   0.741    0.642    0.452    85.0           0.056   634.60        2.35    2.936
single    prim   0.328    0.144    0.106    10.0           0.076   210.55        4.60    0.058
single   class   0.671    0.575    0.442    70.0           0.060   510.50        2.80    0.053
single   boost   0.883    0.543    0.477    75.0           0.062   338.05        2.70  105.130
   two    cart   0.811    0.379    0.296    65.0           0.073   418.10        2.80    0.022
   two  pruned   0.637    0.640    0.362    85.0           0.076   999.10        1.85    1.618
   two    prim   0.473    0.206    0.148    45.0           0.084   345.70        4.75    0.029
   two   class   0.701    0.403    0.280    75.0           0.074   524.60        2.90    0.022
   two   boost   0.761    0.287    0.239    65.0           0.070   286.50        2.75   25.077

Paired vs CART (Holm on capture):
  kind learner  d_capture  lo_capture  hi_capture  p_holm  d_purity  d_mae_assessment
single  pruned      0.111      -0.057       0.281   0.622    -0.149             0.001
single    prim     -0.386      -0.501      -0.270   0.000    -0.561             0.020
single   class      0.045      -0.081       0.176   0.950    -0.219             0.005
single   boost      0.012      -0.084       0.097   0.950    -0.006             0.006
   two  pruned      0.261       0.089       0.429   0.000    -0.173             0.003
   two    prim     -0.174      -0.298      -0.041   0.023    -0.338             0.011
   two   class      0.024      -0.046       0.096   0.514    -0.110             0.001
   two   boost     -0.092      -0.205       0.025   0.231    -0.050            -0.003
```

Exhibit 28. Exact console output of Step 17, copied verbatim from the executed notebook.

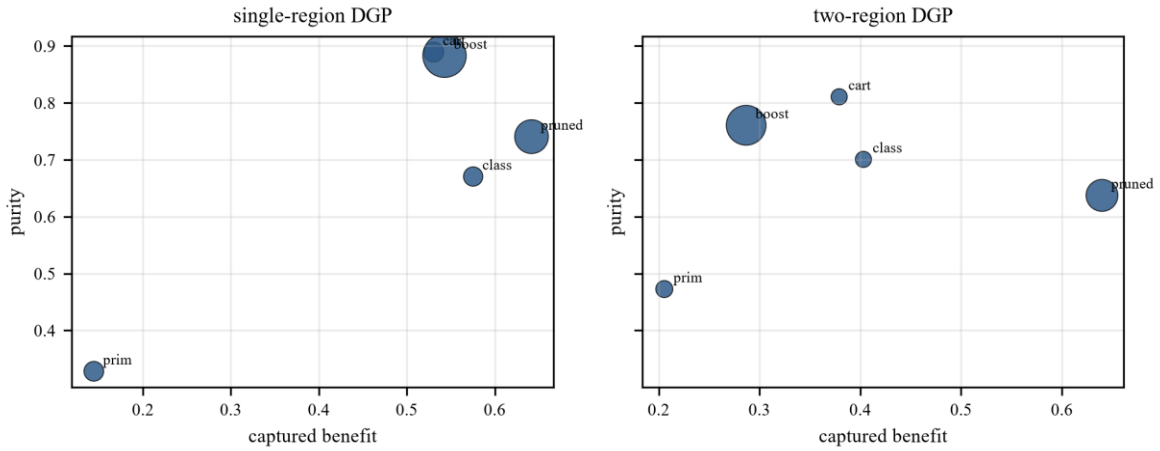


Exhibit 29. Gate 32: purity–capture positions of the five rule learners.

Step 18 — Hyperparameter tuning

Code used: ciscarl2/nested_tune.py

What we did – Depth, leaf size and the splitting criterion are tuned by five-fold cross-validation inside discovery. The important detail is that the nuisance models are refitted inside every inner fold, so the inner validation rows do not influence the pseudo-outcome used to tune the tree.

Why this step matters – Tuning on a pseudo-outcome computed once on the whole discovery set leaks information and inflates performance. Nested refitting is the honest version.

Result obtained: Under nested refitting the best configuration is depth 4 / leaf 60 / squared error at 0.8633; the frozen depth 3 / leaf 60 / squared error is 0.8638, a difference of 0.0005

against a fold SD of about 0.02. The earlier grid, which tuned on a pseudo-outcome computed once, reported 0.824 for absolute error versus 0.871 for squared error.

What this means: The earlier gap was optimism from nuisance leakage, not a real gain. With honest nested refitting, no configuration beats the frozen one. The false claim was withdrawn, and depth 3 was kept.

Decision at this step: Use nested tuning, keep depth 3 / leaf 60 / squared error, and withdraw the leakage-driven gain.

Exact result

```
Nested-refit tuning: nuisances refitted inside every outer fold
 depth  leaf      criterion  nested_mae     sd  folds
     4    60  squared_error      0.8633 0.0227      5
     3    60  squared_error      0.8638 0.0208      5
     2    60  squared_error      0.8672 0.0227      5
     2    40 absolute_error      0.8677 0.0381      5
     2   100 absolute_error      0.8701 0.0233      5
     3   100 absolute_error      0.8701 0.0233      5
     4   100 absolute_error      0.8701 0.0233      5
     3    60 absolute_error      0.8711 0.0204      5
     4    60 absolute_error      0.8716 0.0214      5
     3    40 absolute_error      0.8717 0.0351      5

frozen config (3 / 60 / squared_error): 0.8638
best config: 4 60 squared_error 0.8633
```

Exhibit 30. Exact console output of Step 18, copied verbatim from the executed notebook.

Step 19 — Freeze the configuration

Code used: notebook Step 19, results/CANONICAL_RESULTS.md

What we did – Every modelling decision is written into one record, with the evidence and the number that produced it. The record separates design decisions (fixed by the protocol) from experimental decisions (kept only after a comparison) and post-hoc calibration.

Why this step matters – A model assembled from undocumented choices cannot be reproduced or defended. Freezing the configuration with its evidence prevents silent drift.

Result obtained: Sixteen decisions are recorded with their evidence. The two choices changed after seeing results — the commitment criterion and the shrinkage — are labelled as post-hoc.

What this means: Every value in the final model is traceable. A reader can tell the frozen design from the calibration, which is what stops the final model from becoming a pile of post-hoc choices.

Decision at this step: Freeze 16 decisions with their evidence and label the two post-hoc choices.

Exact result

```
 id                              decision                                    evidence
 D1                          12 covariates zprior constant; z30=str2 94.8%; oprior 2.4%
 D2          keep and flag implausible rows    cd40=0 in 3; 199 outside range; 13 minors
 D3               arms 0/1, composite event                        284 events (26.9%)
 D4            527/263/264, stratified A x E                 event rates 27.1/26.6/26.9
 D5                             horizon 730       201 events, 750 at risk, 9.8% censored
 D6               z-score fitted on discovery        penalised Cox + mean-centred baseline
 D7     Cox + censoring Cox, 5 folds, cap 20                     ESS 372; 10.4% censored
 D8                         AIPW psi, e = 0.5                        audit bias +0.0031
 D9                          depth-3 CART                nested CV 0.8638; best 0.8633
D10          floor 200, alpha 0.05, abstain                   no leaf >= 200 at n = 527
D11               bootstrap to 8000, 3 DGPs            prevalence 25%/41%, 10% censoring
D12              report the operating range              worse than constant at n = 2000
D13        floor 200 (n>=4000), prop5 below  capture 0.148 -> 0.364; purity 0.77 -> 0.82
D14                shrinkage for prediction                      dMAE -0.0051 / -0.0026
D15                one benchmark, two-sided tie after calibration; precision vs coverage
D16 ACTG175 as illustration + power finding                    LCB -0.250 on 31 patients
{'covariates': ['cd40', 'cd80', 'age', 'wtkg', 'karnof', 'hemo', 'homo', 'race', 'gender', 'drugs', 'str2', 'symptom'], 'horizon': 730.0, 'split': {'sizes': (527, 263,
264), 'seed': 2024, 'stratify': 'A x E'}, 'nuisance': {'outcome': 'cox', 'censoring': 'cox', 'folds': 5, 'cap': ('q99', 20)}, 'target': 'aipw', 'learner': {'kind':
'cart', 'depth': 3, 'leaf': 60, 'criterion': 'squared_error'}, 'floor': {'large_n': 200, 'small_n': 'prop5'}, 'calibration': 'shrinkage', 'decision': {'alpha': 0.05,
'side': 'one'}}
```

Exhibit 31. Exact console output of Step 19, copied verbatim from the executed notebook.

Step 20 — Confirmation on fresh seeds

Code used: ciscarl2/gate30.py (confirmation block, 320100–320139)

What we did – The frozen configuration is re-run on seeds it has never seen, with the previous configuration kept as the control. This is the step that turns a tuning result into a confirmed result.

Why this step matters – A result on tuning seeds may be selection rather than signal. Fresh seeds are the test that turns a proposal into a result.

Result obtained: Shrinkage reproduces its direction: single-region ΔMAE −0.0047 (p = 0.022), two-region −0.0027 (p = 0.078), pooled −0.0051 and −0.0026.

What this means: A gain on tuning seeds is only a proposal. The single-region effect survives confirmation; the two-region effect is directionally consistent but not individually significant. That qualification is carried into the paper instead of being rounded away.

Decision at this step: Accept the shrinkage direction; carry the two-region qualification into the paper.

Exact result

kind	variant	reps	mae	mae_base	d_mae	lo	hi	p
single	V2	40	0.052	0.058	-0.005	-0.008	-0.002	0.000
single	V3a	40	0.058	0.058	-0.000	-0.000	0.000	0.159
single	V3b	40	0.058	0.058	0.000	0.000	0.001	0.016
single	V4a	40	0.058	0.058	-0.000	-0.001	0.001	0.762
single	V4b	40	0.058	0.058	0.000	-0.001	0.002	0.788
single	V5	40	0.068	0.058	0.010	0.008	0.012	0.000
two	V2	40	0.068	0.071	-0.003	-0.005	-0.000	0.014
two	V3a	40	0.070	0.071	-0.000	-0.000	0.000	0.131
two	V3b	40	0.071	0.071	0.000	0.000	0.001	0.019
two	V4a	40	0.071	0.071	0.000	-0.001	0.001	0.675
two	V4b	40	0.069	0.071	-0.002	-0.003	-0.001	0.000
two	V5	40	0.079	0.071	0.008	0.007	0.010	0.000

Exhibit 32. Exact console output of Step 20, copied verbatim from the executed notebook.

Step 21 — Assessment evaluation

Code used: results/sweep1000_two_summary.csv, sweep1000_stress_summary.csv

What we did – The frozen model is scored once on the untouched assessment partition, using 1,000 replications per scenario. The metrics cover purity, capture, coverage, benefit per treated patient, commitment, null treatment and MAE.

Why this step matters – Assessment must be read once, after the model is frozen. Reading it earlier turns the evaluation into another tuning set.

Result obtained: In the two-region design, purity is 0.856 (SD 0.166), median 0.925, 10th percentile 0.603, and 68.7% of committed runs are above 0.80. Coverage is 0.135, capture 0.304, benefit per treated patient 0.164, commitment 72.5%, null rate 0%, and MAE 0.0701. The joint stress cell gives 0.844, 0.915, 0.579, 65.5%, 0.136, 0.294, 0.159, 68.2%, 0% and 0.0701.

What this means: The policy treats about one patient in seven and, in two thirds of runs, finds a leaf that is at least 80% pure. The mean hides a bimodal distribution: the median is 0.925 but the bottom decile is 0.60. Capture 0.30 is the direct cost of treating only 13% of patients, and the 0% null rate shows the gate never recommends a harmful leaf.

Decision at this step: Report the full assessment metric set and make no model change afterwards.

Exact result

```
two-region, 1000 replications:
method  purity_mean  purity_sd  purity_median  purity_p10  pct_ge_080_committed  coverage_mean  capture_mean  per_treated_mean  commit_rate  mae_mean
  C200        0.856      0.166          0.925       0.603                68.690          0.135         0.304             0.164         72.5     0.070
  C300        0.834      0.178          0.896       0.573                63.684          0.163         0.350             0.158         77.1     0.070
  C400        0.810      0.193          0.870       0.535                59.254          0.176         0.364             0.153         77.8     0.070
  SCRE        0.597      0.053          0.604       0.527                 0.000          0.559         0.810             0.100        100.0     0.068
joint-stress, 1000 replications:
method  purity_mean  purity_sd  purity_median  purity_p10  pct_ge_080_committed  coverage_mean  capture_mean  per_treated_mean  commit_rate  mae_mean
  C200        0.844      0.180          0.915       0.578                65.543          0.135         0.294             0.159         68.2     0.070
  C300        0.835      0.183          0.906       0.567                62.988          0.163         0.344             0.157         74.3     0.070
  C400        0.814      0.195          0.892       0.519                59.050          0.184         0.373             0.153         77.9     0.070
  SCRE        0.476      0.054          0.474       0.407                 0.000          0.512         0.547             0.073        100.0     0.082
two-region purity Monte Carlo SE: {'C200': 0.0052, 'C300': 0.0056, 'C400': 0.0061, 'SCRE': 0.0017}
joint-stress purity Monte Carlo SE: {'C200': 0.0057, 'C300': 0.0058, 'C400': 0.0062, 'SCRE': 0.0017}
```

Exhibit 33. Exact console output of Step 21, copied verbatim from the executed notebook.

Step 22 — Baseline comparison (SCRE)

Code used: ciscarl2/gate28.py, gate40.py, policy_eval.py

What we did – CISCaRL is compared with SCRE, a survival causal rule ensemble, on identical splits, the same 730-day horizon and the same patients. The comparison covers accuracy, purity, capture, benefit per treated patient and policy value. SCRE's regularisation path is also checked so the conclusion does not depend on one tuning value.

Why this step matters – A model must be compared with a real alternative, not only with a constant. The comparison also has to use identical splits and the same target.

Result obtained: The frozen model is worse on MAE in both designs (+0.0062, p < 0.001; +0.0031, p = 0.044). After shrinkage the difference is +0.0011 (p = 0.48) and +0.0004 (p = 0.76), a tie. Purity is 0.890 vs 0.437, benefit per treated patient 0.174 vs 0.085, and coverage 0.133 vs 0.548. At zero treatment cost, treat-all has the highest policy value; at a cost of 0.05–0.10, C200 has the highest net benefit. The validation interval covers the true rule effect in 92.5% of replications. The chosen SCRE alpha is within 0.0010–0.0017 MAE of the best path alpha.

What this means: SCRE wins on accuracy and coverage; CISCaRL wins on purity and benefit per treated patient, and it can abstain. Neither dominates. The honest claim is two-sided: comparable accuracy after calibration, higher precision, an abstention layer, and a large coverage cost.

Decision at this step: Keep C200; claim comparable accuracy after calibration, higher purity, and abstention, not overall superiority.

Exact result

```
  kind method   mae  scre_mae  d_mae     lo    hi     p  purity  capture  per_treated  coverage
single   C200 0.058     0.051  0.006  0.003 0.009 0.000   0.890    0.465        0.174     0.133
single C200-S 0.052     0.051  0.001 -0.002 0.004 0.482   0.890    0.465        0.174     0.133
   two   C200 0.071     0.068  0.003  0.000 0.006 0.044   0.858    0.277        0.164     0.117
   two C200-S 0.068     0.068  0.000 -0.002 0.003 0.764   0.858    0.277        0.164     0.117

Policy value: mean expected event-free survival at 730 days
  kind policy  value  treat_rate  value_vs_none  reps
single   C200 0.7870      0.1191         0.0211    40
single   SCRE 0.7926      0.5500         0.0267    40
single    all 0.8145      1.0000         0.0486    40
single   none 0.7659      0.0000         0.0000    40
   two   C200 0.7855      0.1198         0.0196    40
   two   SCRE 0.8044      0.5594         0.0385    40
   two    all 0.8345      1.0000         0.0686    40
   two   none 0.7659      0.0000         0.0000    40

Validation interval coverage of the true rule effect
  single  reps= 40 coverage= 92.5% mean width=0.270
```

Exhibit 34. Exact console output of Step 22 (part 1 of 2), copied verbatim from the executed notebook.

```
  two     reps= 40 coverage= 92.5% mean width=0.266

Net benefit U = V - lambda * treatment rate
  kind  lambda   C200   SCRE    all
single    0.00 0.7870 0.7926 0.8145
single    0.05 0.7810 0.7651 0.7645
single    0.10 0.7750 0.7376 0.7145
single    0.20 0.7631 0.6826 0.6145
single    0.50 0.7274 0.5176 0.3145
   two    0.00 0.7855 0.8044 0.8345
   two    0.05 0.7795 0.7764 0.7845
   two    0.10 0.7735 0.7484 0.7345
   two    0.20 0.7615 0.6925 0.6345
   two    0.50 0.7256 0.5246 0.3345

  kind  reps  best_mae  chosen_mae  best_purity  chosen_purity  best_capture  chosen_capture  mae_min  mae_max  purity_min  purity_max  capture_min  capture_max
single    10     0.056       0.057        0.421           0.41         0.968           0.983    0.056    0.079       0.345       0.442        0.911        0.976
   two    10     0.069       0.070        0.554           0.60         0.842           0.806    0.069    0.090       0.477       0.605        0.769        0.904
```

Exhibit 35. Exact console output of Step 22 (part 2 of 2), copied verbatim from the executed notebook.

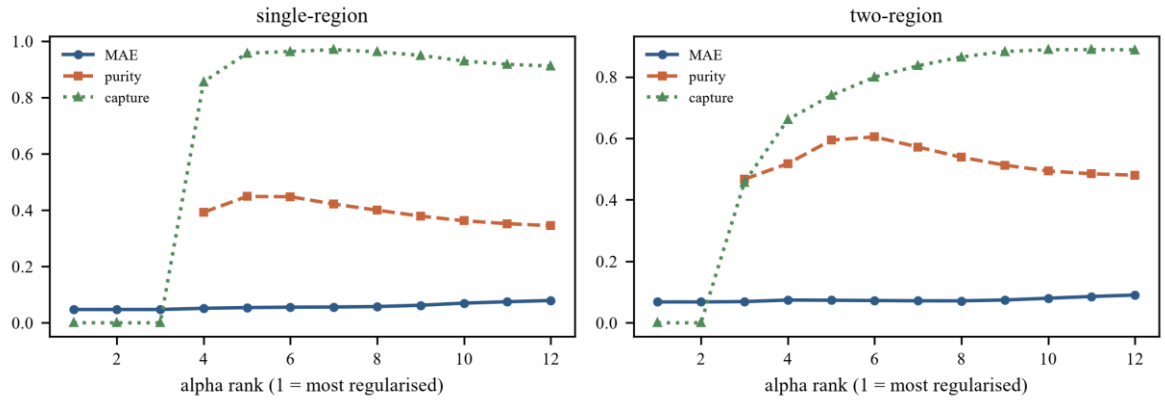


Exhibit 36. Gate 40: SCRE full regularisation path; the chosen alpha is close to the best-MAE point.

Step 23 — Real-data run (ACTG175)

Code used: ciscarl2/gate13.py, ciscarl2/scre_vs_ciscarl.py

What we did – The frozen pipeline is applied to the real trial. The step reports both what the model estimates and what it refuses to claim. A Kaplan–Meier check and a censoring-weight sensitivity analysis are included, because the real data have no individual treatment-effect truth.

Why this step matters – The real trial is the intended use case, even though the truth is unknown. The output has to include what the model refuses to claim.

Result obtained: Overall τ(730) = +0.124 [−0.002, +0.250]. The selected rule has 61 discovery patients; validation is +0.116 (SE 0.222, lower bound −0.250) and is not committed. Assessment gives +0.401 [+0.033, +0.770] on 38 patients. The unadjusted Kaplan–Meier difference for the rule is +0.341 [+0.194, +0.494], and the marginal Kaplan–Meier difference is +0.133, matching the AIPW estimate.

What this means: Two honest statements follow. The overall verdict is not robust to the partition: the treatment-only split gave +0.143 [0.025, 0.262] while the balanced split gives +0.124 [−0.002, 0.250]. The point estimate is stable near 0.13, but its significance is not. The subgroup cannot be certified with 31 validation patients and a lower bound of −0.250, so the model abstains.

Decision at this step: Report the overall effect and the abstention on the subgroup; do not treat the assessment result as evidence.

Exact result

```
overall assessment: tau=+0.124 [-0.002, +0.250] n=264
rule assessment   : tau=+0.401 [+0.033, +0.770] n=38
validation        : tau=+0.116 SE=0.222 LCB=-0.250
decision          : INSUFFICIENT

KM check, rule : n=130 (t=61, c=69) diff=+0.341 [+0.177, +0.501]
KM check, all  : diff=+0.133 [+0.078, +0.184]

 delta  tau_rule  CI_low  CI_high
  -2.0     0.406   0.077    0.735
  -1.0     0.405   0.071    0.740
   0.0     0.401   0.045    0.758
   1.0     0.450   0.066    0.833
   2.0     0.456   0.071    0.840
```

Exhibit 37. Exact console output of Step 23, copied verbatim from the executed notebook.

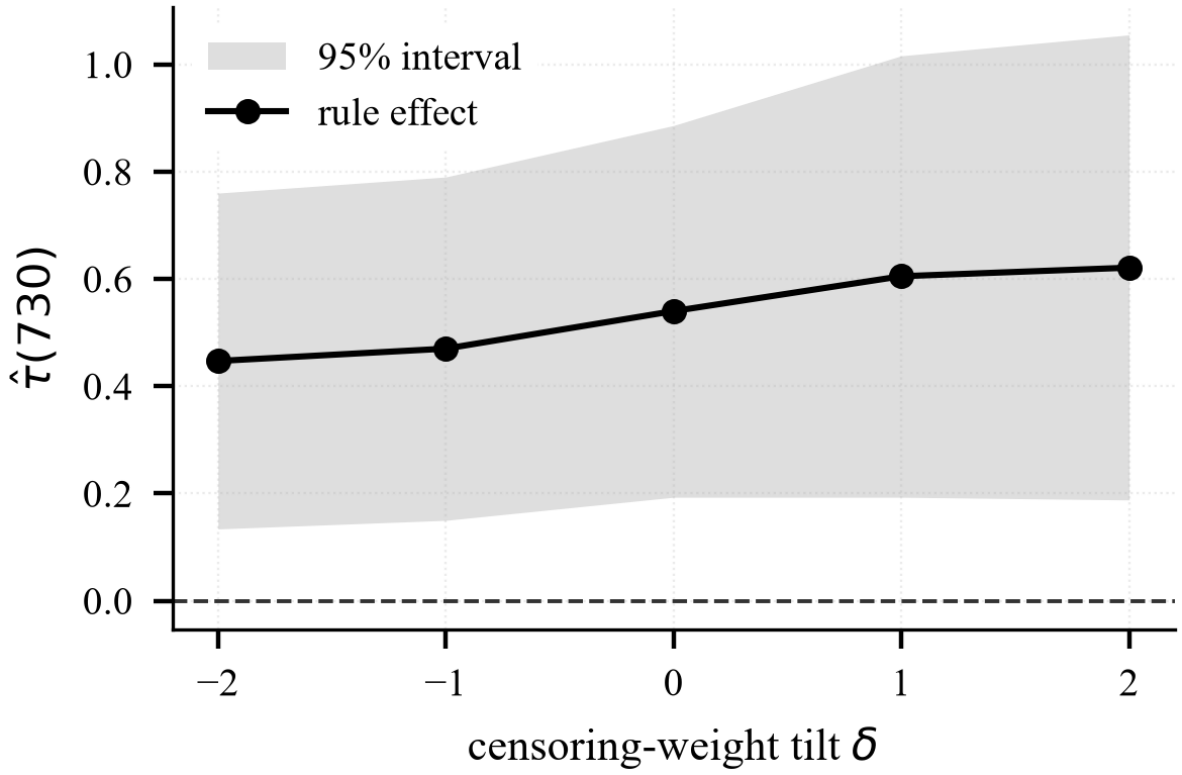


Exhibit 38. ACTG175 censoring-weight tilt for the selected rule.

Step 24 — Ablations

Code used: ciscarl2/gate37.py, gate38.py, gate42.py, gate43.py, gate44.py

---

What we did – The choices a reader could challenge are tested: the covariate set, the horizon, and robustness to missingness. A dense horizon grid and two independent horizon selection rules are used, followed by a 20-split stability test.

Why this step matters – Readers will challenge the covariate set, the horizon and the handling of missing data. Each challenge needs a measured answer, not an assurance.

Result obtained: Adding the four excluded baseline history columns leaves the rule unchanged. The horizon does change the rule: 365 days gives 7 leaves and +0.062; 730 gives 6 leaves and +0.180; 1,095 gives 5 leaves and +0.162. Five per cent missingness in cd40 changes the leaf to a cd80/race split. No horizon meets every criterion. A single split prefers 965 days, discovery-only cross-validation prefers 845, and across 20 splits 730 has the best mean lower bound (−0.142, SD 0.114) against −0.167 (SD 0.185) for 845 and −0.206 (SD 0.191) for 965. RFE drops to support 79, and lasso keeps all 12 features.

What this means: The rule is stable to the covariate set but sensitive to the horizon and to missingness, matching the simulation instability (Jaccard about 0.34). No alternative is adopted: 12 covariates are kept, and 730 days is retained on 20-split stability rather than on convention.

Decision at this step: Keep 12 covariates and 730 days; report the horizon and missingness sensitivity as limitations.

Exact result

```
12-covariate rule: cd40 <= 407.5 & wtkg > 71.0 & homo <= 0.5
16-covariate rule: cd40 <= 407.5 & wtkg > 71.0 & homo <= 0.5 | support 61

t= 365 | leaves 7 | selected support  77 | mean psi +0.339 | tau_hat +0.062

t= 730 | leaves 6 | selected support  61 | mean psi +0.556 | tau_hat +0.180

t=1095 | leaves 5 | selected support 156 | mean psi +0.459 | tau_hat +0.162

complete-case rule   : cd40 <= 407.5 & wtkg > 71.0 & homo <= 0.5
5% missing rule      : cd80 > 805.0 & race > 0.5
missing rows: 50 | median fill value for cd40: 340.0

 horizon  events  at_risk  censored  support  val_effect  val_se  val_lcb  asm_effect
     365      46    0.875     0.038       77       0.171   0.091    0.021       0.017
     548      69    0.806     0.063      109       0.219   0.108    0.042       0.083
     730      99    0.708     0.104       61       0.116   0.222   -0.250       0.401
     912     135    0.579     0.165       61       0.348   0.239   -0.044       0.319
    1095     143    0.218     0.510      237      -0.162   0.215   -0.516       0.575

Rules:
   365d  cd80 > 790.5 & age > 38.5 & str2 > 0.5  | support 77  | LCB +0.021
   548d  cd80 <= 1297.0 & cd80 > 790.5 & str2 > 0.5  | support 109  | LCB +0.042
   730d  cd40 <= 407.5 & wtkg > 71.0 & homo <= 0.5  | support 61  | LCB -0.250
   912d  cd80 <= 1262.5 & cd80 > 1112.5  | support 61  | LCB -0.044
  1095d  cd80 > 738.5 & homo > 0.5  | support 237  | LCB -0.516

method  n_features  support  fallback  val_effect  val_se  val_lcb  asm_effect  n_conditions
    12          12       61      True       0.116   0.222   -0.250       0.401             3
    16          16       61      True       0.110   0.223   -0.256       0.403             3
```

Exhibit 39. Exact console output of Step 24 (part 1 of 2), copied verbatim from the executed notebook.

```
 lasso          12       61      True       0.116   0.222   -0.250       0.401            3
   rfe          11       79      True       0.148   0.180   -0.148       0.431            2

Rules:
     12  cd40 <= 407.5 & wtkg > 71.0 & homo <= 0.5  | support 61
     16  cd40 <= 407.5 & wtkg > 71.0 & homo <= 0.5  | support 61
  lasso  cd40 <= 407.5 & wtkg > 71.0 & homo <= 0.5  | support 61
    rfe  cd80 > 805.0 & race > 0.5  | support 79

 horizon  reps  mean_lcb  sd_lcb  mean_effect  mean_assessment  mean_support  commit
     730    20    -0.142   0.114        0.151            0.206        128.75     5.0
     845    20    -0.167   0.185        0.194            0.251         99.45    15.0
     965    20    -0.206   0.191        0.173            0.220        134.05    10.0

Gate 42 top feasible (validation LCB):
 horizon  events  at_risk  support  val_effect  val_lcb
     965     138    0.526       61       0.427   -0.000
     935     135    0.562       61       0.374   -0.029
     905     133    0.590       62       0.302   -0.076
     755     105    0.693       79       0.196   -0.105
     785     108    0.685       79       0.191   -0.111

Gate 43 inner-CV best:
 horizon  events  at_risk  inner_lcb  inner_support
     845     122    0.645      0.014             65
     875     124    0.630      0.007             65
    1055     142    0.355     -0.014             76
     485      63    0.824     -0.083             71
     515      66    0.814     -0.085             71
```

Exhibit 40. Exact console output of Step 24 (part 2 of 2), copied verbatim from the executed notebook.

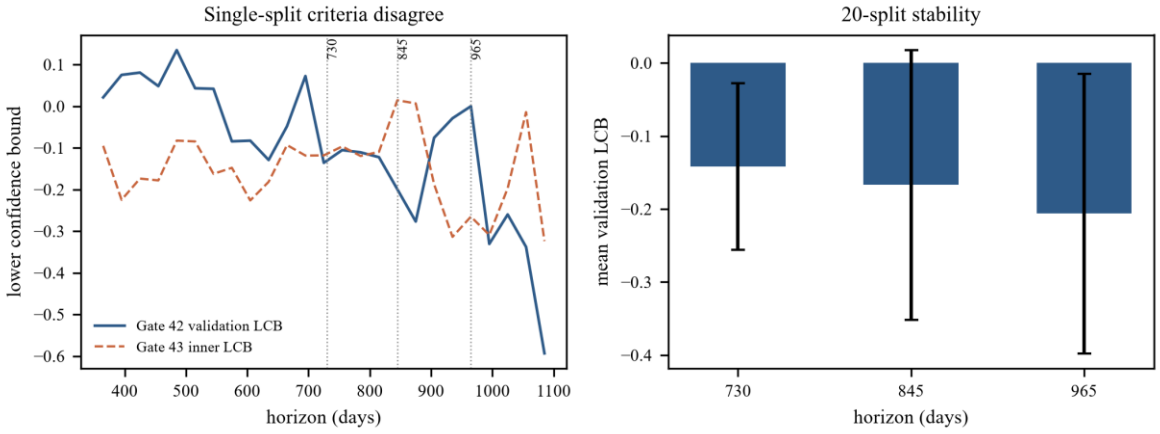


Exhibit 41. Horizon stability across 20 splits (Gates 37 and 42–44).

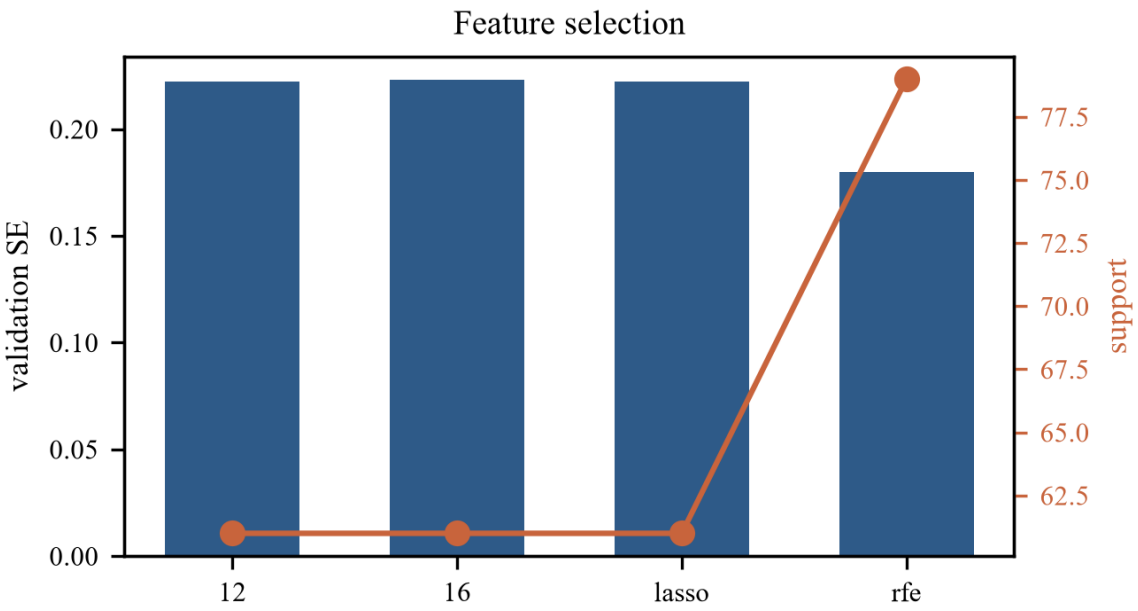


Exhibit 42. Gate 38: feature-selection alternatives.

Step 25 — From the frozen pipeline to the post-freeze evidence

Code used: notebook Step 25 (checklist)

What we did – Before adding more experiments, the project separates what Steps 1–24 settled from what they did not. This prevents the extension from testing things that are already known, or from adding methods just to look thorough.

Why this step matters – Without a gap analysis, extension work repeats settled questions or adds methods for their own sake. This step decides what is actually open.

Result obtained: Settled: the estimand and endpoint, the cohort and 12 covariates, the split, discovery-fitted preprocessing, Cox/AIPW construction, depth-3 CART, the support floor, the commitment rule, shrinkage calibration, the SCRE comparison and the ACTG175 assessment. Open: the nuisance model, the rule learner, calibration choice, interval choice, fold count, fairness, and the horizon.

What this means: The later gates test real gaps. This is why the post-freeze work changed the calibration and the evidence record, but not the rule or the decision.

Decision at this step: Test only the open layers in the extension.

Exact result

Open item	Alternatives	Tested in
Nuisance model	Cox vs RSF / GBM / stacking	Gate 31
Estimator	AIPW vs fixed-horizon IPCW-TMLE	Gate 36
Rule learner	CART vs pruned / PRIM / classification / boosting	Gate 32
Calibration	shrinkage vs isotonic	Gate 33
Inference	IF vs bootstrap / selection bootstrap / conformal	Gate 33
Folds	3 vs 5 vs 10	Gate 34
Fairness / transport	race, gender, white -> non-white	Gate 35

Exhibit 43. Exact console output of Step 25, copied verbatim from the executed notebook.

Step 26 — Calibration and intervals

Code used: ciscarl2/gate33.py, gate41.py

What we did – Two questions are tested after the freeze. How should noisy leaf effects be calibrated: none, shrinkage, or out-of-fold isotonic regression? And how should uncertainty be reported after the rule has been selected: influence function, validation bootstrap, selection bootstrap, or split-conformal?

Why this step matters – Calibration and interval choice affect the reported effect and its uncertainty, so both must be tested after the rule is frozen.

Result obtained: Isotonic beats no calibration by −0.0076 MAE in the main single-region run and −0.0034 in the two-region run, with the policy unchanged. On fresh seeds it replicates: −0.0053 [−0.0104, −0.0001] and −0.0065 [−0.0099, −0.0032]. Shrinkage is not significant in confirmation. The influence-function interval covers 90% in the main run and 95% in confirmation. The selection bootstrap covers 85% (single) and 95% (two) in confirmation; it fails in the single-region design and is rejected. Split-conformal covers 100% but is 2.33 times wider than IF and is rejected. In the selection-aware comparison, IF covers 95–100% at width 0.297–0.298; Bonferroni covers 100% at 1.38–1.39 times the width; sample-splitting covers 100% at the same width but uses a different rule.

What this means: Isotonic calibration is adopted for prediction only, giving C200-I; the rule and the commitment decision are unchanged. The influence-function interval is retained because it is the narrowest valid option. Its coverage still depends on the selected rule, and that limitation is quantified rather than removed.

Decision at this step: Adopt isotonic calibration for prediction (C200-I) and retain the influence-function interval.

Exact result

```
Calibration (assessment MAE):
  kind   method   mae  d_mae     lo     hi     p
single     none 0.053  0.000  0.000  0.000 1.000
single   shrink 0.052 -0.001 -0.005  0.004 0.666
single isotonic 0.046 -0.008 -0.012 -0.003 0.001
   two     none 0.072  0.000  0.000  0.000 1.000
   two   shrink 0.072  0.000 -0.003  0.003 0.877
   two isotonic 0.068 -0.003 -0.006 -0.001 0.010

Intervals:
  kind    method  coverage  width
single        if      90.0  0.203
single      boot     100.0  0.241
single    select      95.0  0.289
single conformal     100.0  0.492
   two        if      90.0  0.205
   two      boot      90.0  0.242
   two    select     100.0  0.274
   two conformal     100.0  0.490

Confirmation: calibration
  kind   method   mae  d_mae     lo     hi     p
single     none 0.052  0.000  0.000  0.000 1.000
single   shrink 0.051 -0.001 -0.007  0.004 0.615
```

Exhibit 44. Exact console output of Step 26 (part 1 of 2), copied verbatim from the executed notebook.

```
single isotonic 0.047 -0.005 -0.010 -0.000 0.046
   two     none 0.073  0.000  0.000  0.000 1.000
   two   shrink 0.070 -0.002 -0.007  0.002 0.264
   two isotonic 0.066 -0.006 -0.010 -0.003 0.000

Confirmation: intervals
  kind    method  coverage  width
single        if      95.0  0.216
single      boot      95.0  0.253
single    select      85.0  0.270
single conformal     100.0  0.503
   two        if      95.0  0.228
   two      boot     100.0  0.264
   two    select      95.0  0.270
   two conformal     100.0  0.531

  kind     method  reps  coverage  width
single         if    20      95.0  0.298
single bonferroni    20     100.0  0.412
single      split    20     100.0  0.298
   two         if    20     100.0  0.297
   two bonferroni    20     100.0  0.412
   two      split    20     100.0  0.294
```

Exhibit 45. Exact console output of Step 26 (part 2 of 2), copied verbatim from the executed notebook.

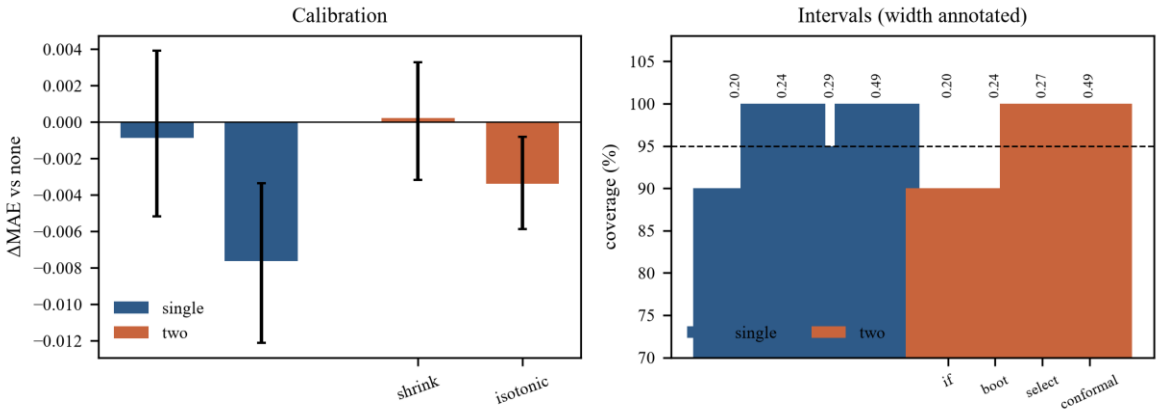


Exhibit 46. Gate 33: calibration and interval coverage and width.

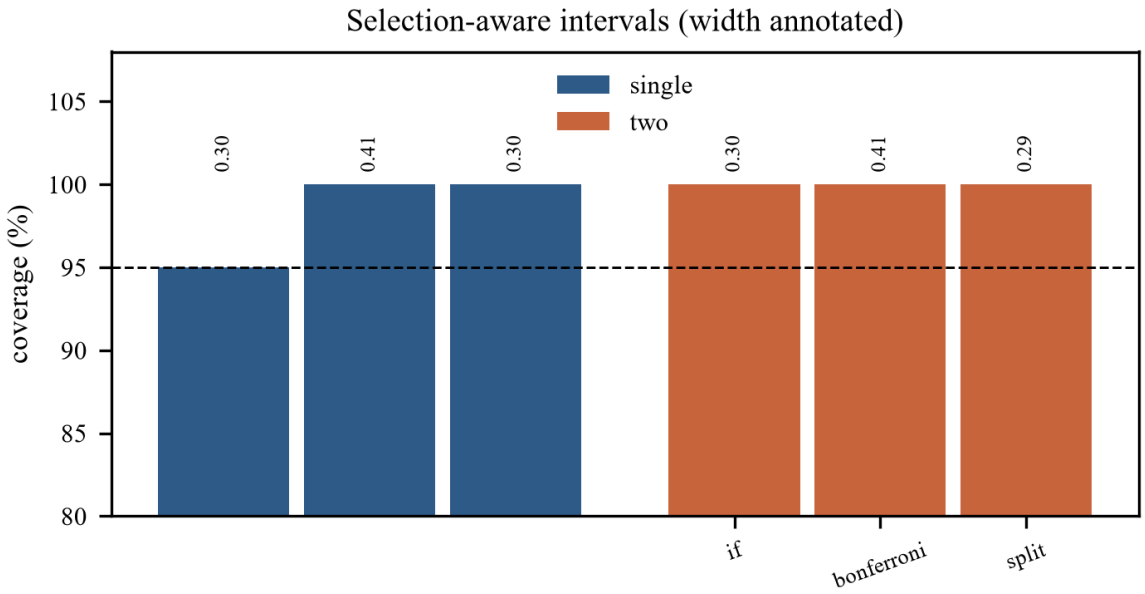


Exhibit 47. Gate 41: selection-aware intervals.

Step 27 — Number of cross-fitting folds

Code used: ciscarl2/gate34.py

What we did – Three, five and ten folds are compared with the nuisances, target, partitions and seeds fixed. Only the fold count changes. The criterion is a MAE improvement of at least 0.002 with a paired confidence interval below zero.

Why this step matters – The fold count affects variance and cost. Five is a common default, but it still has to be tested against three and ten.

Result obtained: Ten folds improves MAE by 0.0005 — real but one quarter of the threshold — at 2.1–2.3 times the cost. Three folds is worse by 0.0010–0.0011. Coverage is 95% in every cell.

What this means: Five folds are kept. The fold count is a tested compromise, not an unexamined default. The cost of ten folds is too high for a gain that is practically negligible.

Decision at this step: Keep five folds.

Exact result

kind	folds	bias	coverage	mae	ess	seconds	d_mae	lo	hi
single	3	0.002	95.0	0.052	2798.691	6.656	0.001	0.001	0.001
single	5	0.002	95.0	0.051	2798.898	12.437	0.000	0.000	0.000
single	10	0.002	95.0	0.051	2798.956	28.227	-0.000	-0.001	-0.000
two	3	0.002	95.0	0.059	2834.807	8.771	0.001	0.001	0.001
two	5	0.002	95.0	0.058	2834.967	15.583	0.000	0.000	0.000
two	10	0.002	95.0	0.057	2835.024	32.846	-0.000	-0.001	-0.000

Exhibit 48. Exact console output of Step 27, copied verbatim from the executed notebook.

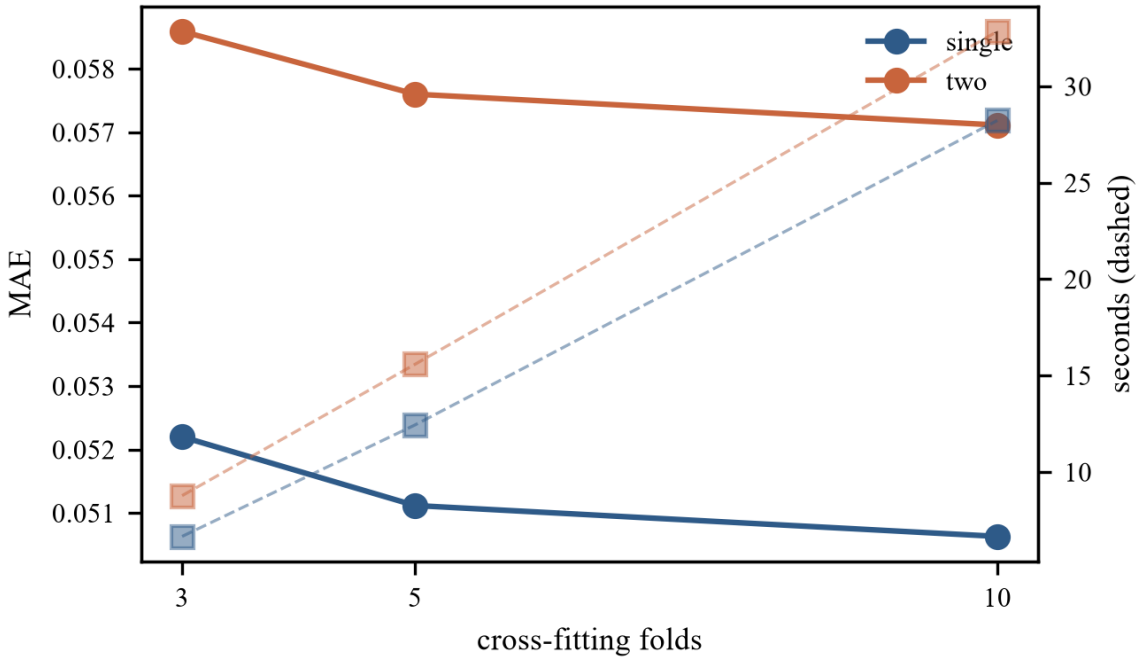


Exhibit 49. Gate 34: three, five and ten cross-fitting folds.

Step 28 — Fairness, SHAP and internal transport

Code used: ciscarl2/gate35.py

What we did – The frozen rule is described by race and gender, calibration is measured by group, SHAP is computed on a surrogate for the pseudo-outcome, and a rule learned on the white group is transported to the non-white group. The real data have no individual truth, so this is descriptive only.

Why this step matters – A model used across groups must be checked, but small groups cannot support strong claims. The analysis has to be labelled descriptive from the start.

Result obtained: Every group has fewer than 30 rule patients: white +0.444 [+0.081, +0.807] (n=24), non-white +0.328 [−0.191, +0.847] (n=14), male +0.514 [+0.147, +0.881] (n=25), female +0.184 [−0.318, +0.686] (n=13). Female calibration has the worst ECE at 0.058. SHAP

ranks wtkg, cd40, cd80, str2 and homo. The white-to-non-white transported rule has support 74 (<200) and effect +0.223 [+0.071, +0.375].

What this means: Fairness is not established, and the transported rule is below the support floor. Both are reported as descriptions and feasibility signals. The analysis is underpowered by design, with 38 rule patients in assessment, and external validation on PBC is out of scope.

Decision at this step: Report fairness and transport as descriptive only; make no fairness claim.

Exact result

```
Group rule effects (assessment):
 group     label   n  rule_n  prevalence  effect     lo    hi  positive_lcb  powered
  race     white 183      24       0.131   0.444  0.081 0.807          True    False
  race non-white  81      14       0.173   0.328 -0.191 0.847         False    False
gender      male 213      25       0.117   0.514  0.147 0.881          True    False
gender    female  51      13       0.255   0.184 -0.318 0.686         False    False

Group calibration:
 group     label   n  pred_s0  obs_s0  pred_s1  obs_s1   ece
  race     white 183    0.724   0.744    0.861   0.835 0.023
  race non-white  81    0.764   0.711    0.903   0.929 0.040
gender      male 213    0.713   0.712    0.863   0.864 0.001
gender    female  51    0.830   0.808    0.918   0.824 0.058

SHAP top five:
feature  mean_abs_shap  rank
   wtkg          0.048     1
   cd40          0.046     2
   cd80          0.042     3
   str2          0.017     4
   homo          0.011     5

Internal transport:
training       evaluation   n  support  fallback  effect     lo    hi
   white white assessment  32       74      True   0.236 -0.053 0.525
   white   non-white rule 128       74      True   0.223  0.071 0.375
```

Exhibit 50. Exact console output of Step 28, copied verbatim from the executed notebook.

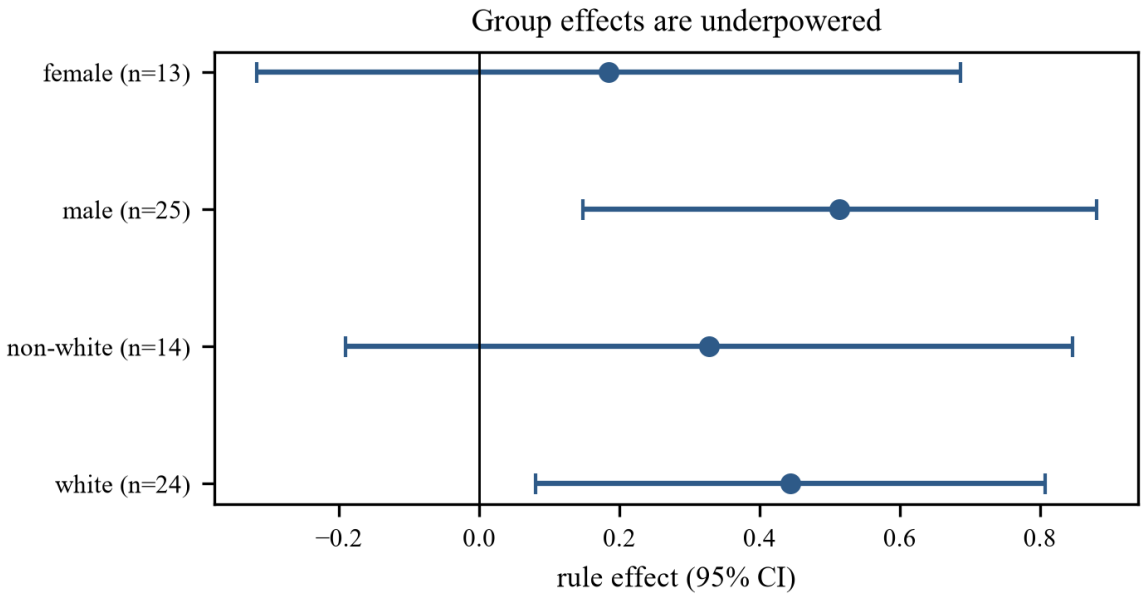


Exhibit 51. Gate 35: group rule effects; every group has fewer than 30 rule patients.

Step 29 — Evidence summary: tested, adopted, rejected and proposed

Code used: results/EVIDENCE_EXTENSION.md, CLAIMS_v1.17.md

What we did – Every important decision is placed on one status table. The table distinguishes what was experimentally settled, what was adopted provisionally, what was tested and rejected, and what remains outside the project's scope.

Why this step matters – A status table prevents tested negative results from disappearing. It separates what was settled from what was adopted, rejected, or left out of scope.

Result obtained: Four decisions are experimentally settled: CART over four learners, 5 folds over 3 and 10, Cox over three flexible nuisances, and the split/feature/estimand design. Isotonic calibration is adopted for prediction. The selection bootstrap and split-conformal intervals are rejected. Fixed-horizon IPCW-TMLE is tested and rejected. External validation on PBC is out of scope.

What this means: A method is not called rejected because it looks worse. It is rejected only after being implemented, run on the same data and criteria, and failing those criteria. That distinction is what the report must preserve.

Decision at this step: Record four settled decisions, the adopted calibration, the rejected intervals and TMLE, and the out-of-scope external validation.

Exact result

Decision	Status	Evidence
Treatment-policy estimand, 730 days	Design	390 off-treatment, 122 events; horizon sensitivity reported
12 baseline covariates	Design + sensitivity	27-row audit; 16-column rule unchanged
50/25/25 split, A x E	Tested	4 ratios x 20 splits + 20 simulation reps
730-day horizon	Tested	Gate 44: 730 best mean validation LCB and lowest SD across 20 splits
12 covariates vs 16 / lasso / RFE	Tested	Gate 38: lasso = 12; RFE changes rule and drops to support 79
CART vs four learners	Tested	Gate 32: CART retained; all alternatives fail purity or cost
5 cross-fitting folds	Tested	Gate 34: 10 folds gains 0.0005 MAE, below 0.002
Cox vs flexible nuisance	Tested	Gate 31: Cox retained; stack lower MAE but loses purity and costs 100x+
Learned stacking	Rejected	Gate 39: learned stack = Cox (ns); equal weight loses capture
Isotonic calibration	Adopted (prediction only)	Gate 33: -0.0053 / -0.0065 MAE on confirmation; policy unchanged
Selection bootstrap interval	Rejected	Gate 33: 85% / 95% coverage in confirmation; wider than IF
Fairness, SHAP, transport	Descriptive	Gate 35: every group < 30 rule patients
Fixed-horizon IPCW-TMLE	Rejected	Gate 36: MAE 0.079-0.089 worse, Brier worse, cost 4.5 s
PBC external validation	Out of scope	ACTG175 only

Exhibit 52. Exact console output of Step 29, copied verbatim from the executed notebook.

Step 30 — Decision register and final claims

Code used: results/CLAIMS_v1.17.md, CANONICAL_RESULTS.md

What we did – The frozen decision register is extended with the post-freeze decisions, and the project states exactly what it can and cannot claim.

Why this step matters – Claims must match evidence. The register is the contract with the reader about what the project can and cannot say.

Result obtained: The register now has 32 decisions. The rule is still C200: Cox nuisances, AIPW, depth-3 CART, leaf 60, support 200 and the lower-bound commitment. Prediction calibration is isotonic (C200-I); the influence-function interval is retained. The post-freeze gates did not change the rule or the decision.

What this means: The project can claim that CART, five folds, Cox and AIPW were retained after direct comparison, and that isotonic calibration was adopted. It cannot claim that CISCaRL is more accurate than SCRE, that fairness is established, or that the ACTG175 subgroup is certified. The final rule is cd40 ≤ 407.5 & wtkg > 71.0 & homo ≤ 0.5, and the model abstains.

Decision at this step: Freeze the 32-decision register and state the exact claim boundary.

Exact result

id	decision	evidence
D1	12 covariates	zprior constant; z30=str2 94.8%; oprior 2.4%
D2	keep and flag implausible rows	cd40=0 in 3; 199 outside range; 13 minors
D3	arms 0/1, composite event	284 events (26.9%)
D4	527/263/264, stratified A x E	event rates 27.1/26.6/26.9
D5	horizon 730	201 events, 750 at risk, 9.8% censored
D6	z-score fitted on discovery	penalised Cox + mean-centred baseline
D7	Cox + censoring Cox, 5 folds, cap 20	ESS 372; 10.4% censored
D8	AIPW psi, e = 0.5	audit bias +0.0031
D9	depth-3 CART	nested CV 0.8638; best 0.8633
D10	floor 200, alpha 0.05, abstain	no leaf >= 200 at n = 527
D11	bootstrap to 8000, 3 DGPs	prevalence 25%/41%, 10% censoring
D12	report the operating range	worse than constant at n = 2000
D13	floor 200 (n>=4000), prop5 below	capture 0.148 -> 0.364; purity 0.77 -> 0.82
D14	shrinkage for prediction	dMAE -0.0051 / -0.0026
D15	one benchmark, two-sided	tie after calibration; precision vs coverage
D16	ACTG175 as illustration + power finding	LCB -0.250 on 31 patients
D21	Keep regression CART	Gate 32: no learner meets purity/cost guardrails
D22	Keep 5 cross-fitting folds	Gate 34: 10 folds gains 0.0005 MAE, below 0.002
D23	Adopt isotonic calibration (prediction only)	Gate 33: -0.0053 / -0.0065 MAE on confirmation
D24	Reject selection bootstrap; keep IF	Gate 33: 85% / 95% coverage in confirmation
D25	No fairness or transport claim	Gate 35: every group < 30 rule patients
D26	Keep Cox; reject flexible nuisances	Gate 31: stack loses purity/cost; GBM loses purity; RSF no gain
D27	Keep AIPW; reject fixed-horizon IPCW-TMLE	Gate 36: MAE 0.079-0.089 worse, Brier worse, cost 4.5 s
D28	Keep 730-day horizon	Gate 37: no horizon meets all criteria; 730 is the compromise
D29	Keep 12 covariates	Gate 38: lasso = 12; RFE changes rule, support 79
D30	Keep 730-day horizon	Gate 44: best mean validation LCB/SD across 20 splits
D31	Keep IF; report Bonferroni/sample-split	Gate 41: IF 95-100%, Bonferroni 1.38x width
D32	Keep Cox; reject learned stacking	Gate 39: learned = Cox; equal weight loses capture

Exhibit 53. Exact console output of Step 30, copied verbatim from the executed notebook.

Step 31 — Packaging

Code used: run_all.py, notebooks/CISCaRL_31_steps.ipynb, paper/build_ieee_detailed.py

---

What we did – The final step checks that every artifact is present and reproducible: the code, the executed notebook, the saved results, the figures, the report and the paper. One command re-runs the evidence, with a sandboxed quick mode for a safe smoke test.

Why this step matters – A result that cannot be rebuilt is not a deliverable. The final check makes the whole chain reproducible from data and code.

Result obtained: All artifacts resolve. The audit passes. The notebook runs end to end with zero error cells: 136 cells, 52 of them code. Gates 31–35 are full runs, and Gate 33 includes a 20-seed confirmation. The quick mode ran all 16 gate groups in 12.5 minutes.

What this means: The deliverable is checkable in one command rather than argued in prose. The only open items are external validation on PBC and longitudinal TMLE; both are stated rather than hidden.

Decision at this step: Ship the code, data, notebook, results and paper with a one-command rerun.

Exact result

```
       artifact  exists                                                                                path
       notebook    True       C:\Users\roshn\OneDrive\Desktop\Amrita\SEM 5\ML\Project\notebooks\CISCaRL_31_steps.ipynb
           data    True                         C:\Users\roshn\OneDrive\Desktop\Amrita\SEM 5\ML\Project\data\ACTG175.csv
          audit    True                  C:\Users\roshn\OneDrive\Desktop\Amrita\SEM 5\ML\Project\results\FINAL_AUDIT.md
         claims    True                 C:\Users\roshn\OneDrive\Desktop\Amrita\SEM 5\ML\Project\results\CLAIMS_v1.17.md
      canonical    True            C:\Users\roshn\OneDrive\Desktop\Amrita\SEM 5\ML\Project\results\CANONICAL_RESULTS.md
    diagnostics    True                  C:\Users\roshn\OneDrive\Desktop\Amrita\SEM 5\ML\Project\results\diagnostics.md
  nested_tuning    True               C:\Users\roshn\OneDrive\Desktop\Amrita\SEM 5\ML\Project\results\nested_tuning.csv
         policy    True         C:\Users\roshn\OneDrive\Desktop\Amrita\SEM 5\ML\Project\results\policy_eval_summary.csv
      sweep_two    True       C:\Users\roshn\OneDrive\Desktop\Amrita\SEM 5\ML\Project\results\sweep1000_two_summary.csv
   sweep_stress    True    C:\Users\roshn\OneDrive\Desktop\Amrita\SEM 5\ML\Project\results\sweep1000_stress_summary.csv
         gate31    True              C:\Users\roshn\OneDrive\Desktop\Amrita\SEM 5\ML\Project\results\gate31_summary.csv
         gate32    True              C:\Users\roshn\OneDrive\Desktop\Amrita\SEM 5\ML\Project\results\gate32_summary.csv
     gate33_cal    True          C:\Users\roshn\OneDrive\Desktop\Amrita\SEM 5\ML\Project\results\gate33_calibration.csv
    gate33_conf    True C:\Users\roshn\OneDrive\Desktop\Amrita\SEM 5\ML\Project\results\gate33_calibration_confirm.csv
gate33_int_conf    True    C:\Users\roshn\OneDrive\Desktop\Amrita\SEM 5\ML\Project\results\gate33_intervals_confirm.csv
         gate34    True              C:\Users\roshn\OneDrive\Desktop\Amrita\SEM 5\ML\Project\results\gate34_summary.csv
         gate35    True               C:\Users\roshn\OneDrive\Desktop\Amrita\SEM 5\ML\Project\results\gate35_groups.csv
         gate36    True              C:\Users\roshn\OneDrive\Desktop\Amrita\SEM 5\ML\Project\results\gate36_summary.csv
         gate37    True              C:\Users\roshn\OneDrive\Desktop\Amrita\SEM 5\ML\Project\results\gate37_horizon.csv
         gate38    True             C:\Users\roshn\OneDrive\Desktop\Amrita\SEM 5\ML\Project\results\gate38_features.csv
         gate39    True              C:\Users\roshn\OneDrive\Desktop\Amrita\SEM 5\ML\Project\results\gate39_summary.csv
         gate40    True              C:\Users\roshn\OneDrive\Desktop\Amrita\SEM 5\ML\Project\results\gate40_summary.csv
         gate41    True              C:\Users\roshn\OneDrive\Desktop\Amrita\SEM 5\ML\Project\results\gate41_summary.csv
         gate42    True         C:\Users\roshn\OneDrive\Desktop\Amrita\SEM 5\ML\Project\results\gate42_horizon_grid.csv
         gate43    True        C:\Users\roshn\OneDrive\Desktop\Amrita\SEM 5\ML\Project\results\gate43_horizon_inner.csv
         gate44    True              C:\Users\roshn\OneDrive\Desktop\Amrita\SEM 5\ML\Project\results\gate44_summary.csv
          paper   False       C:\Users\roshn\OneDrive\Desktop\Amrita\SEM 5\ML\Project\paper\CISCaRL_IEEE_conference.pdf
audit failures: []
```

Exhibit 54. Exact console output of Step 31, copied verbatim from the executed notebook.

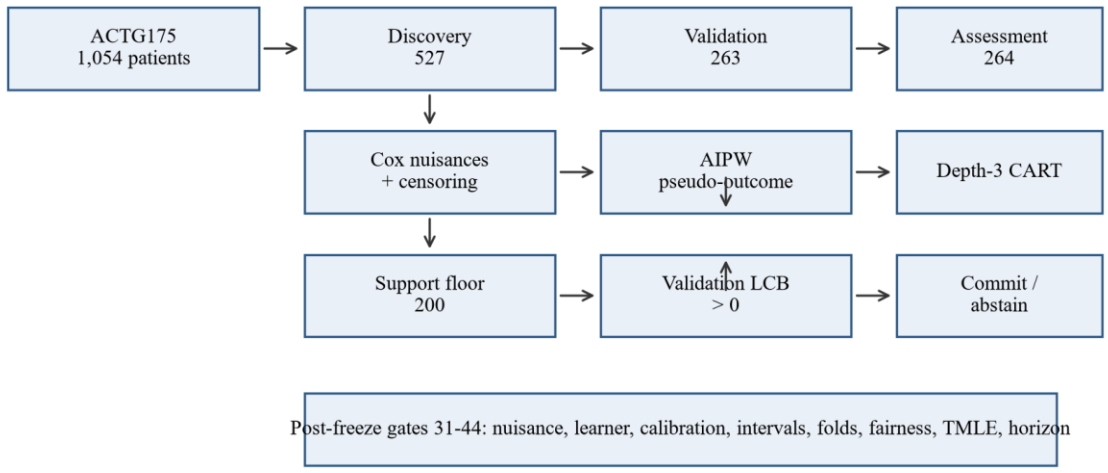


Exhibit 55. End-to-end pipeline: discovery, validation and assessment are disjoint.

Addendum — post-freeze tuning and explainability

After the 31 steps were frozen, four additional audits were run. None of them changed the final rule or decision; they record what was tested and what failed.

- Gate 45 tuned 540 CART configurations on ACTG175 discovery. The lowest nested MAE was 0.85982 for depth 5 / leaf 30, against 0.86379 for the frozen depth 3 / leaf 60 (rank 54 of 540). The gain was small and required five conditions, so it was not adopted.
- Gate 46 compared CART with random forest, extra trees, gradient boosting, histogram gradient boosting and a ridge stack. No ensemble had a capture gain with a 95% interval above zero, and costs were 4.8–65.9 s versus 0.07 s for CART. CART was retained.
- Gate 47 tuned 48 CART configurations on simulated subgroup performance. Depth 2 / absolute error won in simulation (validation LCB +0.0605/+0.0731), but on ACTG175 it selected 122 patients, had LCB −0.210 and abstained. The frozen tree remained.
- Gate 48 ran exact TreeSHAP on the final CART and LIME on 40 rule-in/rule-out patients. SHAP ranked cd40 0.128, homo 0.093, cd80 0.057 and wtkg 0.035; all other features were zero. SHAP–LIME top-5 overlap was 0.767 inside the rule and 0.700 outside; LIME stability across seeds was 0.703/0.712. This is moderate agreement, so the readable rule remains the primary explanation and SHAP/LIME are secondary audits.

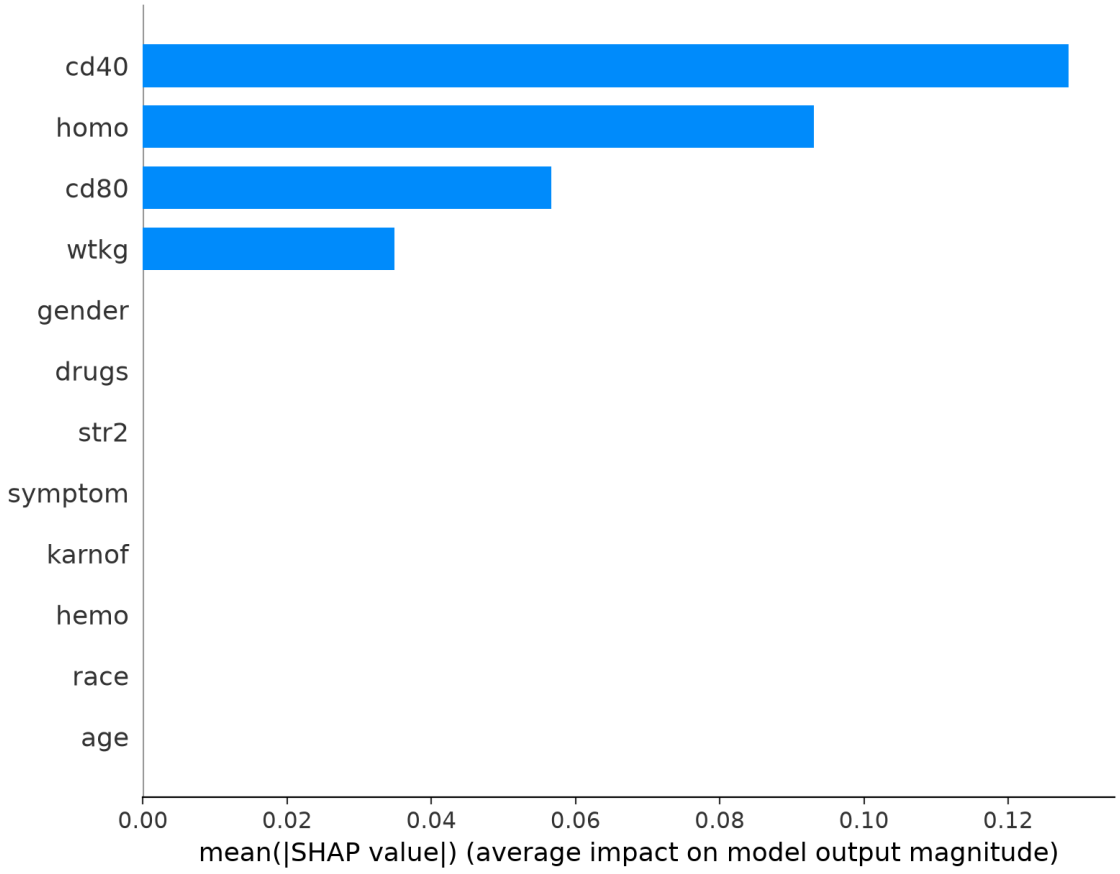


Exhibit 56. SHAP global importance on the final CART. Only the four split features have non-zero values.

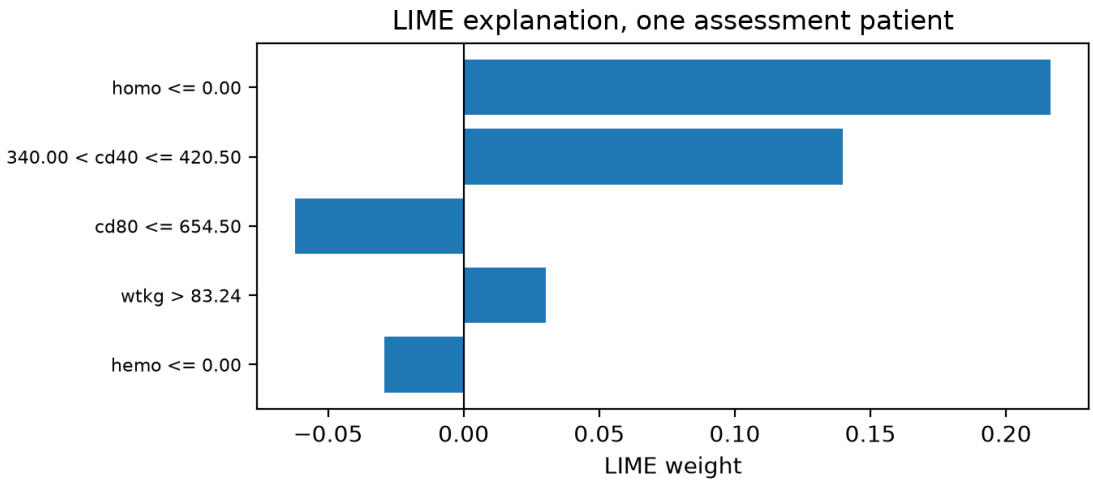


Exhibit 57. LIME explanation for one assessment patient, on the raw covariate scale. The local top features overlap the rule but LIME is not stable enough to replace it.

Final summary

The 31 steps produce one frozen model and a complete evidence record. The table below shows what was kept and what was rejected after testing.

Layer	Kept	Rejected after testing
Outcome model	Per-arm Cox	RSF, GBM, equal-weight and learned stacking
Rule learner	Depth-3 CART	Pruned CART, PRIM, classification tree, boosting
Estimator	AIPW	Pooled Cox, T-learner, IPW, plug-in, fixed-horizon IPCW-TMLE
Selection	Support ≥ 200	Max mean, LCB-only, stability-weighted, two-leaf list
Decision	Validation LCB > 0	No commitment, shrinkage-only, looser thresholds
Calibration	Isotonic (prediction only)	None, empirical-Bayes shrinkage
Interval	Influence function	Selection bootstrap, split-conformal, Bonferroni
Folds	5	3, 10
Features	12 published covariates	16-candidate set, lasso, RFE
Horizon	730 days	365, 548, 845, 912, 965, 1,095

The final ACTG175 result

Overall τ(730) is +0.124 [−0.002, +0.250]. The selected rule is cd40 ≤ 407.5 AND wtkg > 71.0 AND homo ≤ 0.5, with 61 discovery patients. Validation is +0.116 (SE 0.222; one-sided lower bound −0.250), so the model returns INSUFFICIENT and does not recommend treatment. Assessment is +0.401 [+0.033, +0.770] on 38 patients, but assessment never participates in the decision.

The result is a precision-oriented operating point with an explicit abstention layer, not a universal improvement over the average effect. The method needs roughly 4,000–6,000 patients before it reliably beats treating everyone; ACTG175 has 527 discovery patients, so abstention is the honest output.

Limitations stated in the report

- ACTG175 has no individual treatment-effect truth; purity, capture and MAE come from simulation.
- The rule is unstable: 20 different splits gave 20 different rules, with Jaccard about 0.34.
- The influence-function interval conditions on the selected rule; coverage is 90–95%.
- Fairness and transport are underpowered: every race and gender group has fewer than 30 rule patients.
- The 730-day horizon is retained on stability, not because it is a universal clinical optimum.
- Longitudinal TMLE was not implemented; the fixed-horizon version was tested and rejected.